

Lecture 13

Generative Models: Introduction

Speaker: Kaiming He

Overview: Lectures on Generative Models

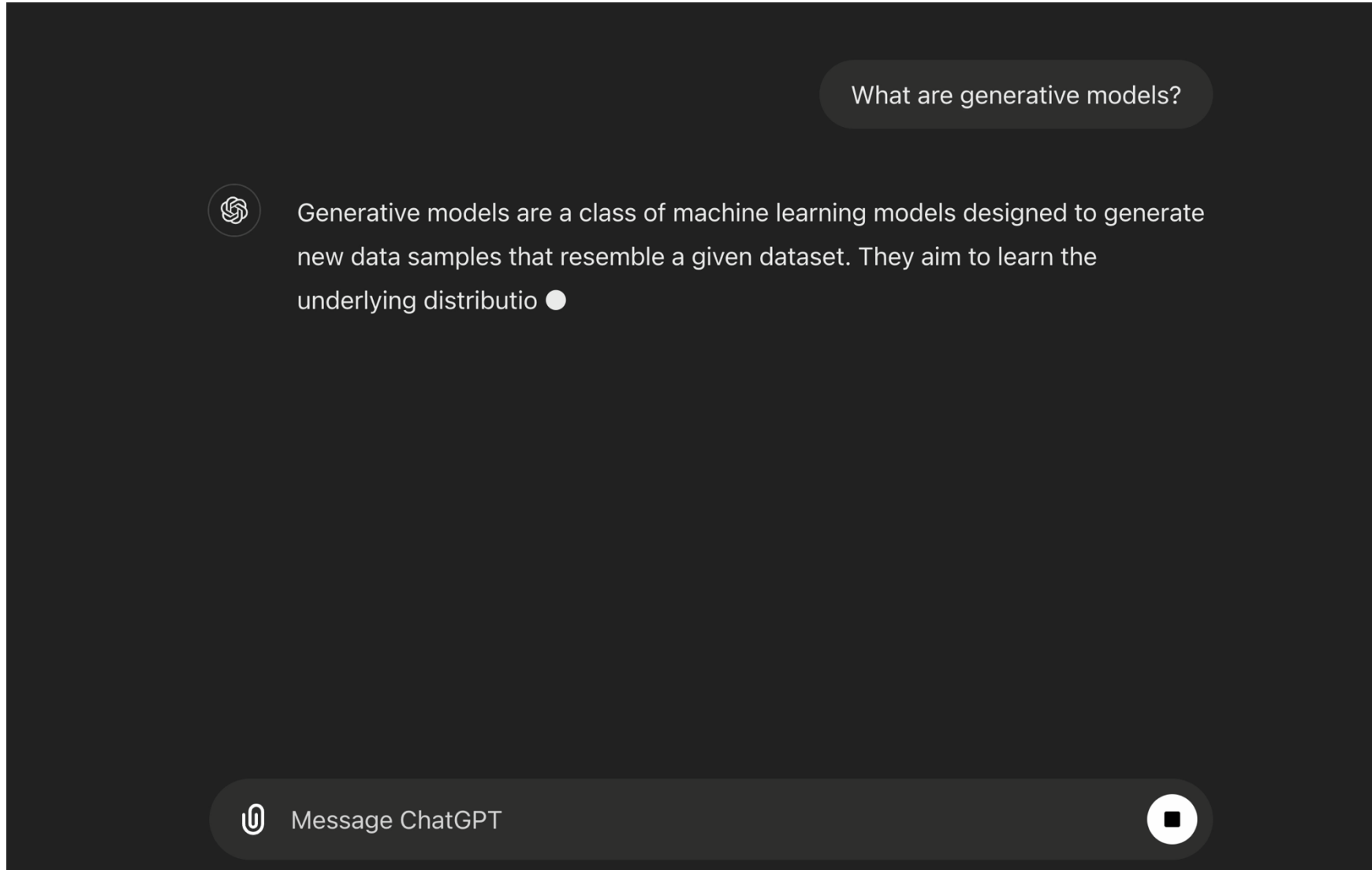
- Today: **Introduction**
- 10/23: **“Classical” methods**
 - Variational Autoencoder (VAE)
 - Generative Adversarial Network (GAN)
- 10/30: **“Modern” methods**
 - Diffusion Models
 - Flow Matching
- Previous lectures: Autoregressive (AR) models - RNN, Transformer, LLM, ...

Today: Introduction to Generative Models

- What are Generative Models?
- Generative Modeling of Real-world Problems
- Families of Generative Models

The “GenAI” Era

Chatbot and natural language conversation



The “GenAI” Era

Text-to-image generation



Generated by Stable Diffusion 3 Medium.

Prompt: teddy bear teaching a course, with "generative models" written on blackboard

The “GenAI” Era

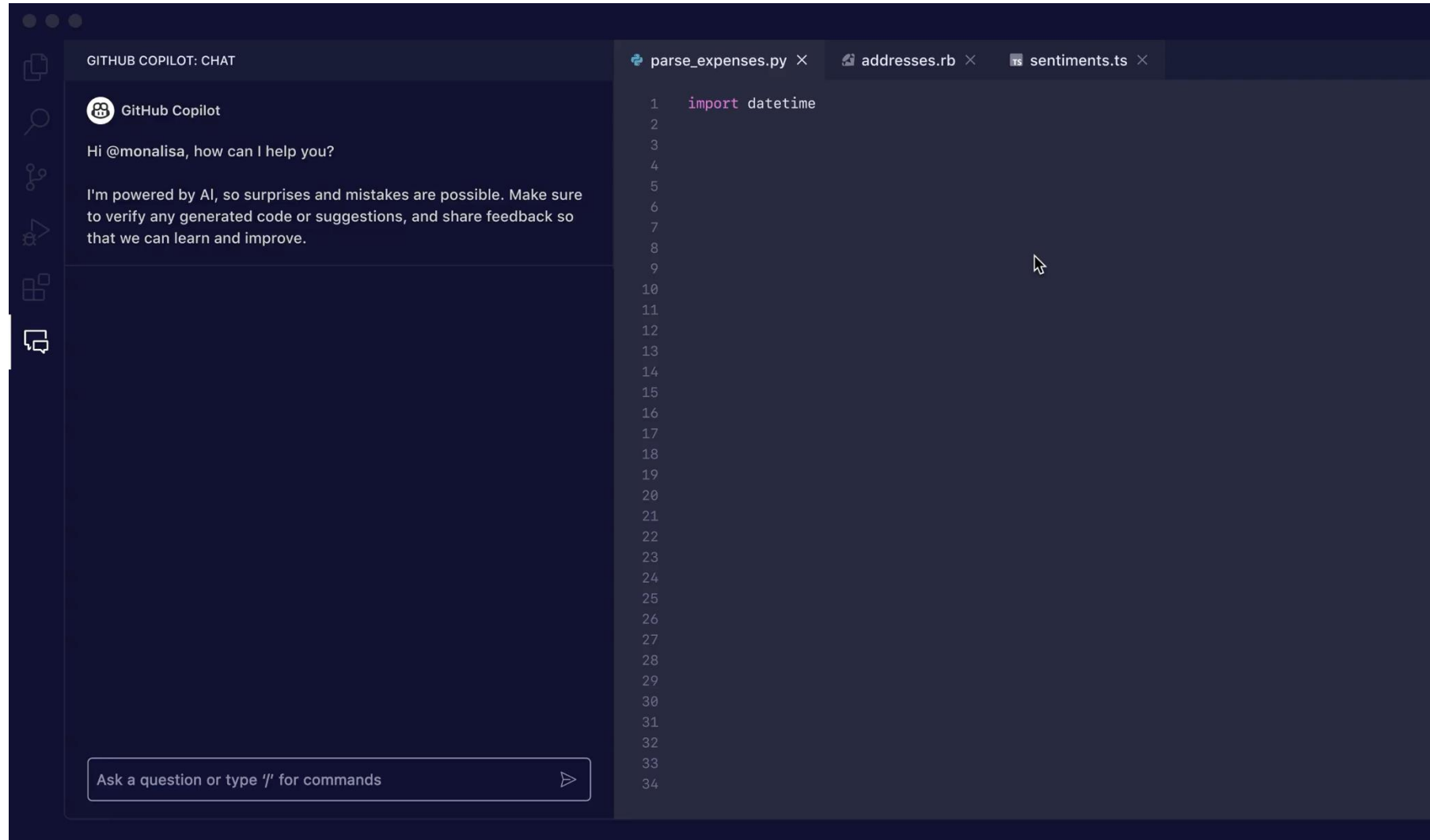
Text-to-video generation



Generated by Sora

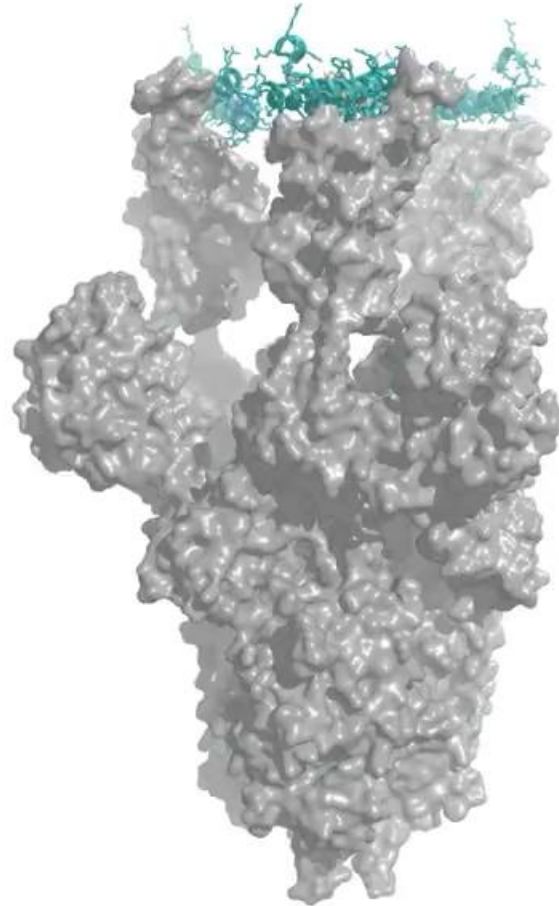
The “GenAI” Era

AI assistant for code generation



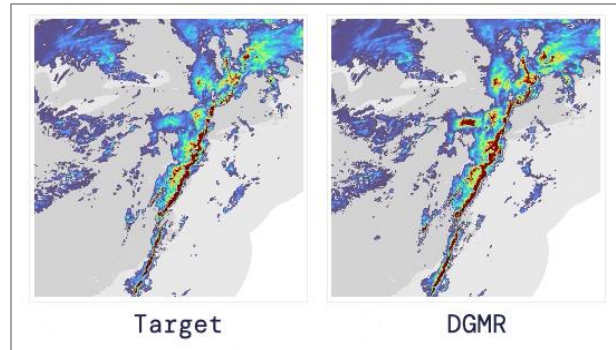
The “GenAI” Era

Protein design and generation

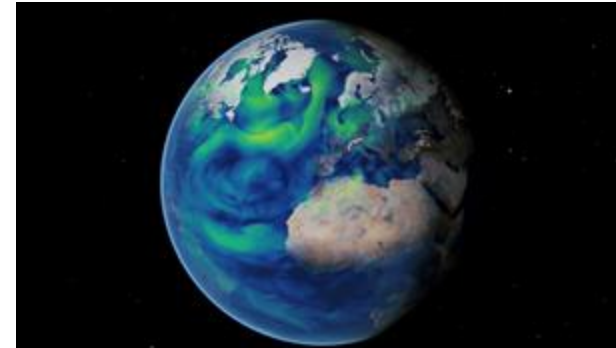


The “GenAI” Era

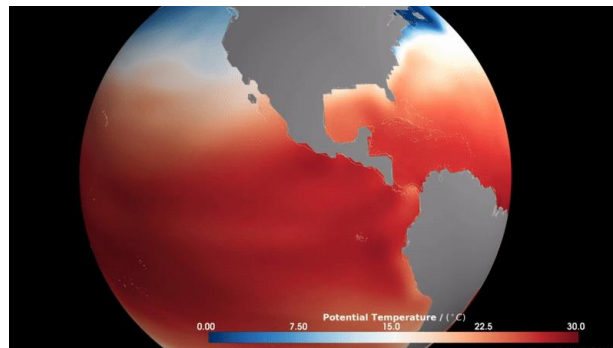
Scientific Problems



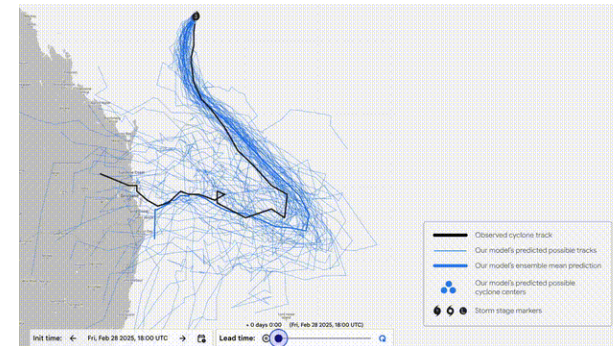
precipitation



weather



ocean



cyclone

Skilful precipitation nowcasting using deep generative models of radar. S Ravuri, et al., Nature, 2021.

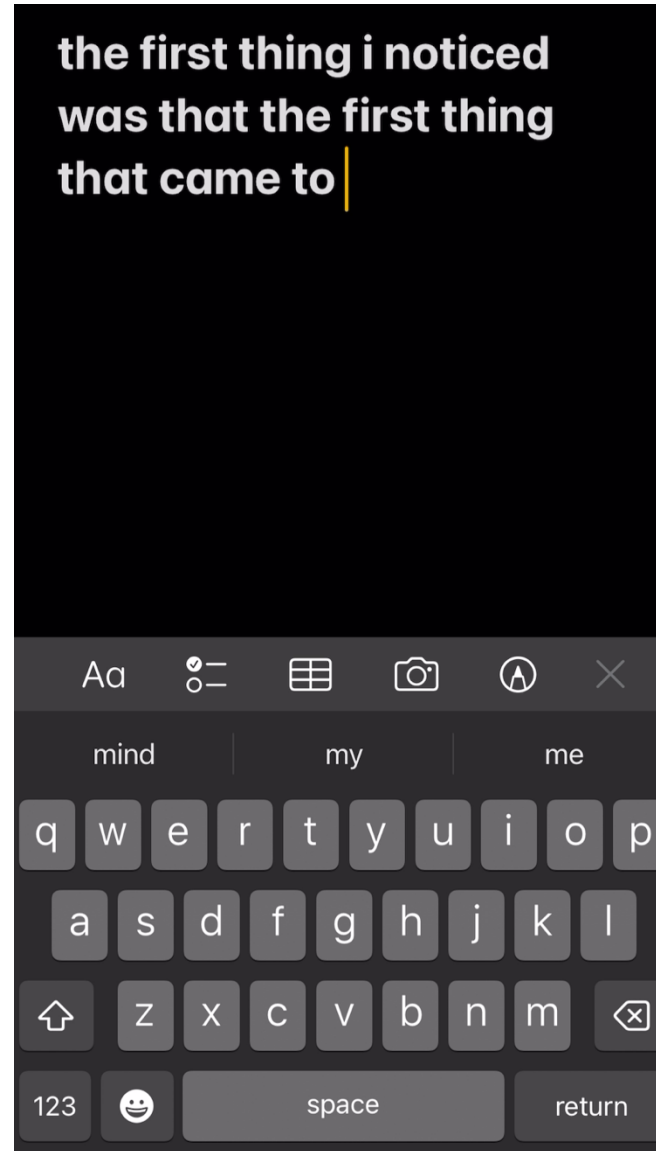
Meet “Samudra”: M²LinES’ AI Global Ocean Emulator for Climate. L Zanna, 2024.

How we're supporting better tropical cyclone prediction with AI. DeepMind, 2025.

FourCastNet 3 Enables Fast and Accurate Large Ensemble Weather Forecasting with Scalable Geometric ML. B Bonev et al., 2025.

Generative Models before the “GenAI” Era

Your keyboard



Generative Models before the “GenAI” Era

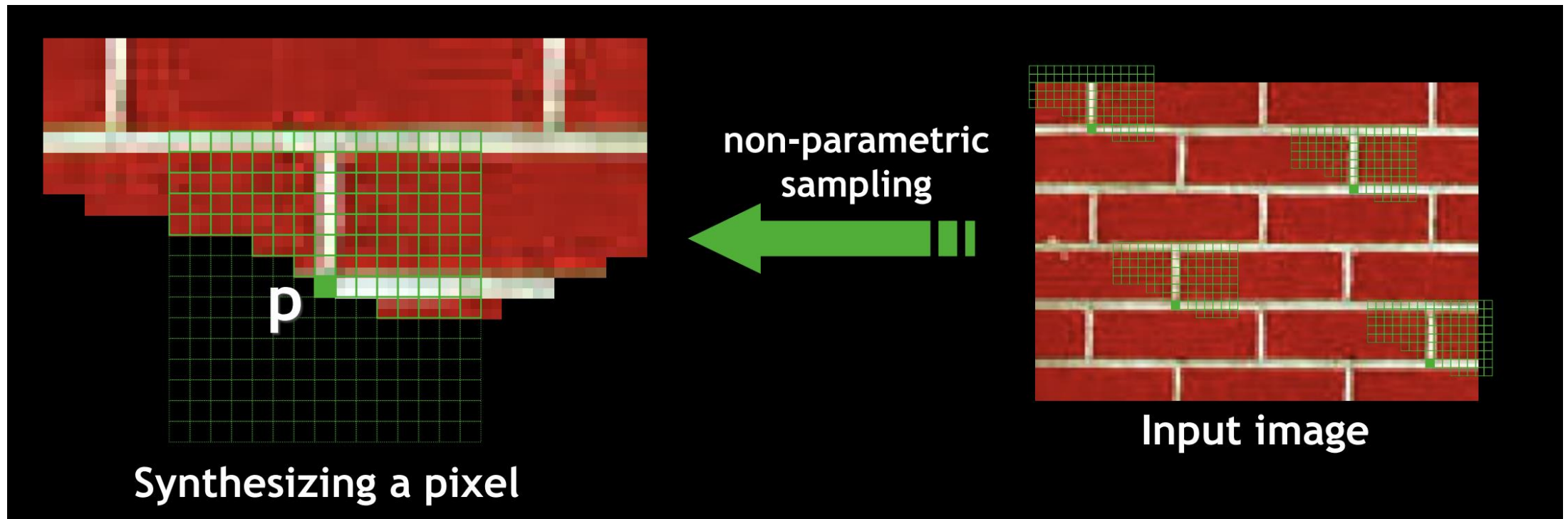
2009, PatchMatch: Photoshop’s Content-aware Fill



Generative Models before the “GenAI” Era

1999, the Efros-Leung algorithm for texture synthesis

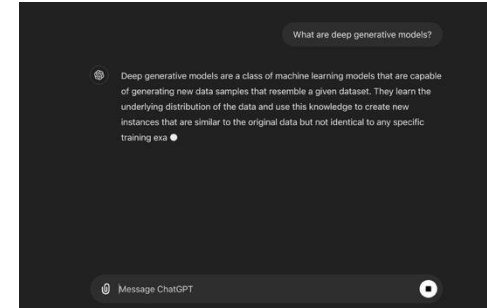
In today's word: this is an **Autoregressive** model



What are Generative Models?

What do these scenarios have in common?

- There are **multiple** predictions to one input.
- Some predictions are more **plausible**.
- Training data may contain **no solution**.
- Output may be **more complex**, more informative, and higher-dimensional



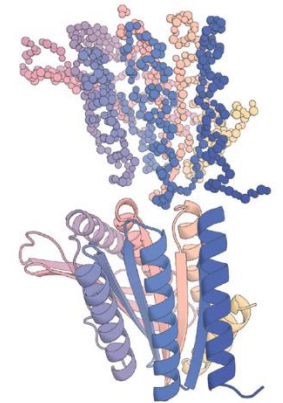
Chatbot



Image generation



Video generation



Protein generation

Discriminative Models vs. Generative Models

Discriminative Model

- “sample” $x \Rightarrow$ “label” y
- one desired output

Generative Model

- “label” $y \Rightarrow$ “sample” x
- many possible outputs

discriminative

x



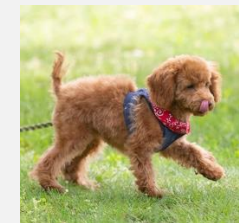
“dog”

y

generative

y

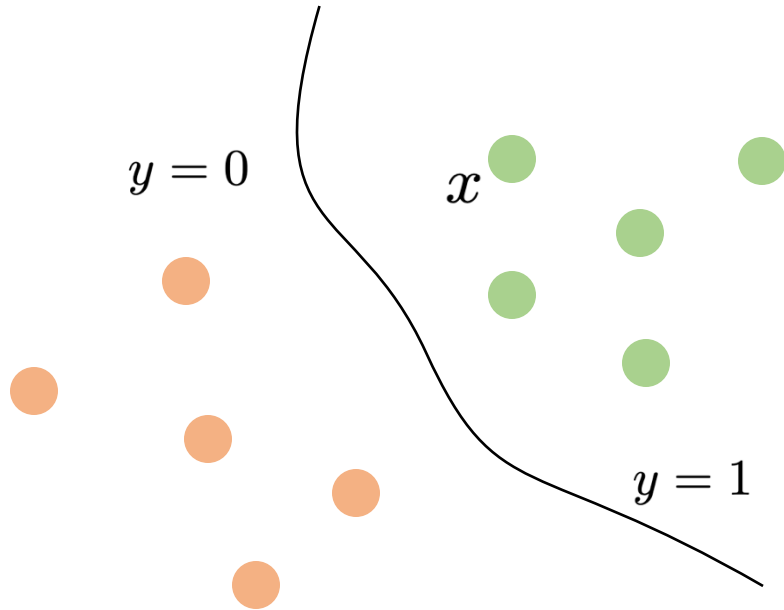
“dog”



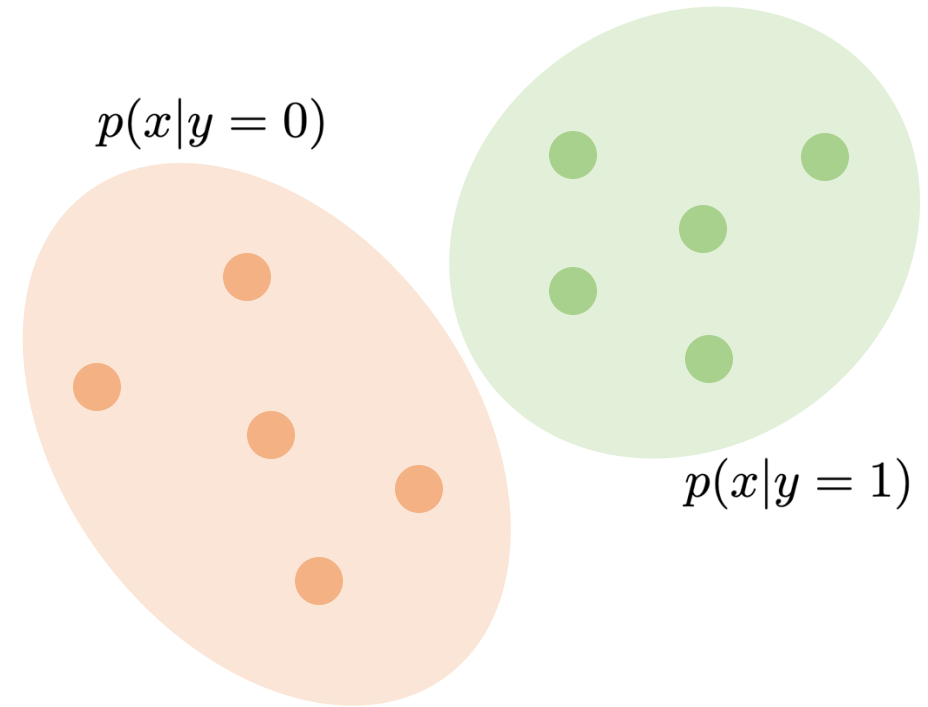
x

Discriminative Models vs. Generative Models

discriminative $p(y|x)$



generative $p(x|y)$



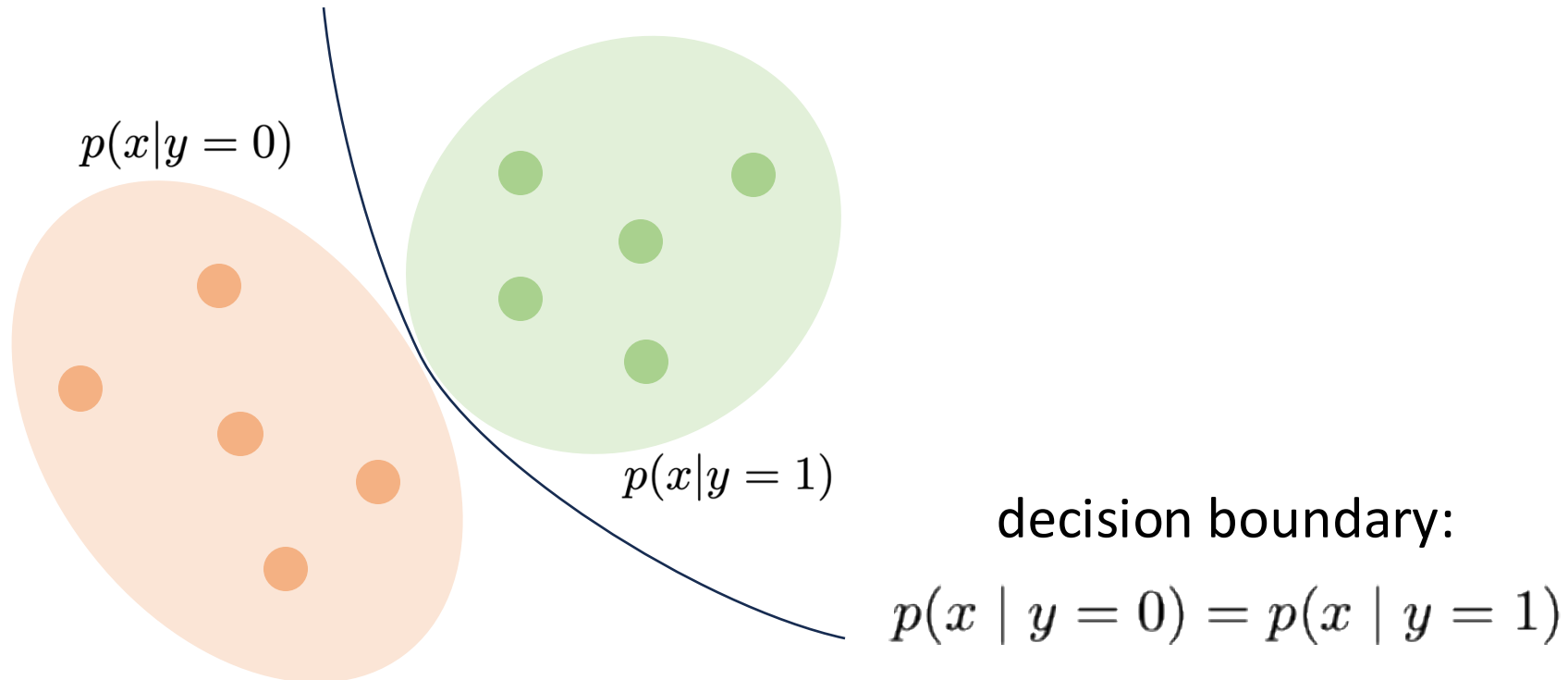
- Generative models can be discriminative: Bayes' rule
- Can discriminative models be generative?

- Generative models can be discriminative: Bayes' rule

$$\underbrace{p(y|x)}_{\text{discriminative}} = \underbrace{p(x|y)}_{\text{generative}} \frac{p(y)}{p(x)}$$

← assuming known prior (uniform categories)

← constant for given x



- Generative models can be discriminative: Bayes' rule

$$\underbrace{p(y|x)}_{\text{discriminative}} = \underbrace{p(x|y)}_{\text{generative}} \frac{p(y)}{p(x)}$$

← assuming known prior (uniform categories)

← constant for given x

- Can discriminative models be generative?

$$\underbrace{p(x|y)}_{\text{generative}} = \underbrace{p(y|x)}_{\text{discriminative}} \frac{p(x)}{p(y)}$$

← **unknown prior distribution of x** ("natural" images?)

← constant for given y

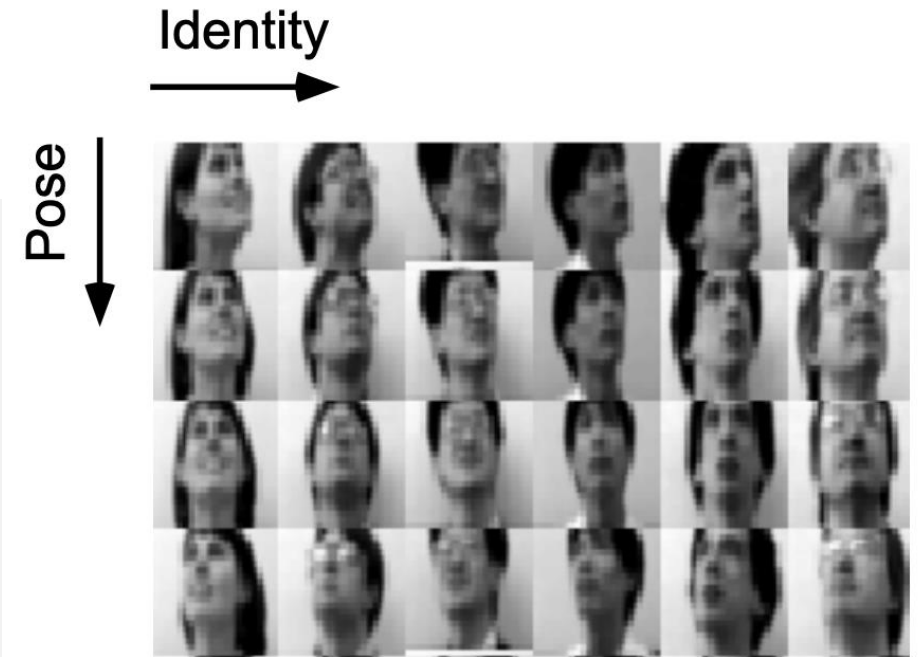
- The challenge is on **probabilistic modeling**

Probabilistic Modeling

- Where does probability come from?
- Assuming underlying **distributions of data generation process**

example:

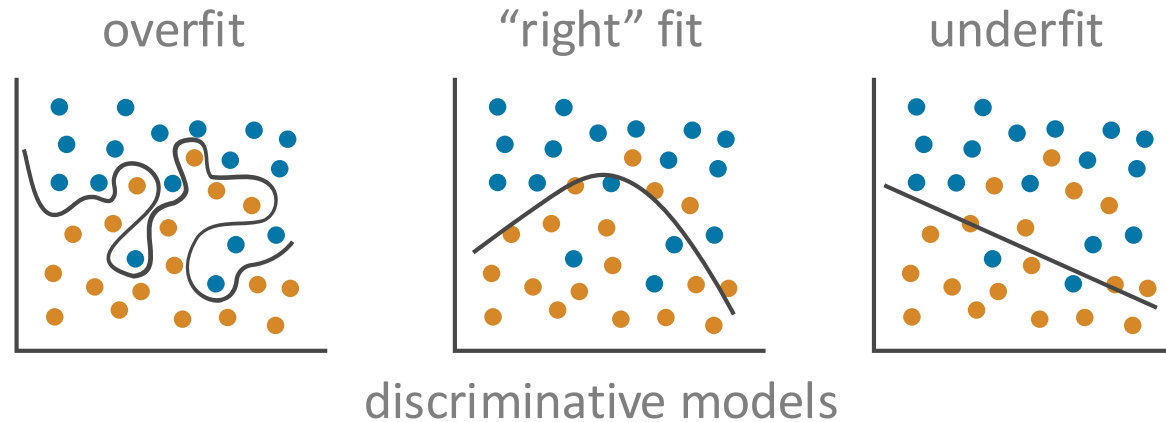
- latent factors z (pose, lighting, scale, ...)
- z has simple distributions
- observations x are rendered by a graphics model, (a function on z)
- x has complex distributions



- Probability itself is a modeling assumption

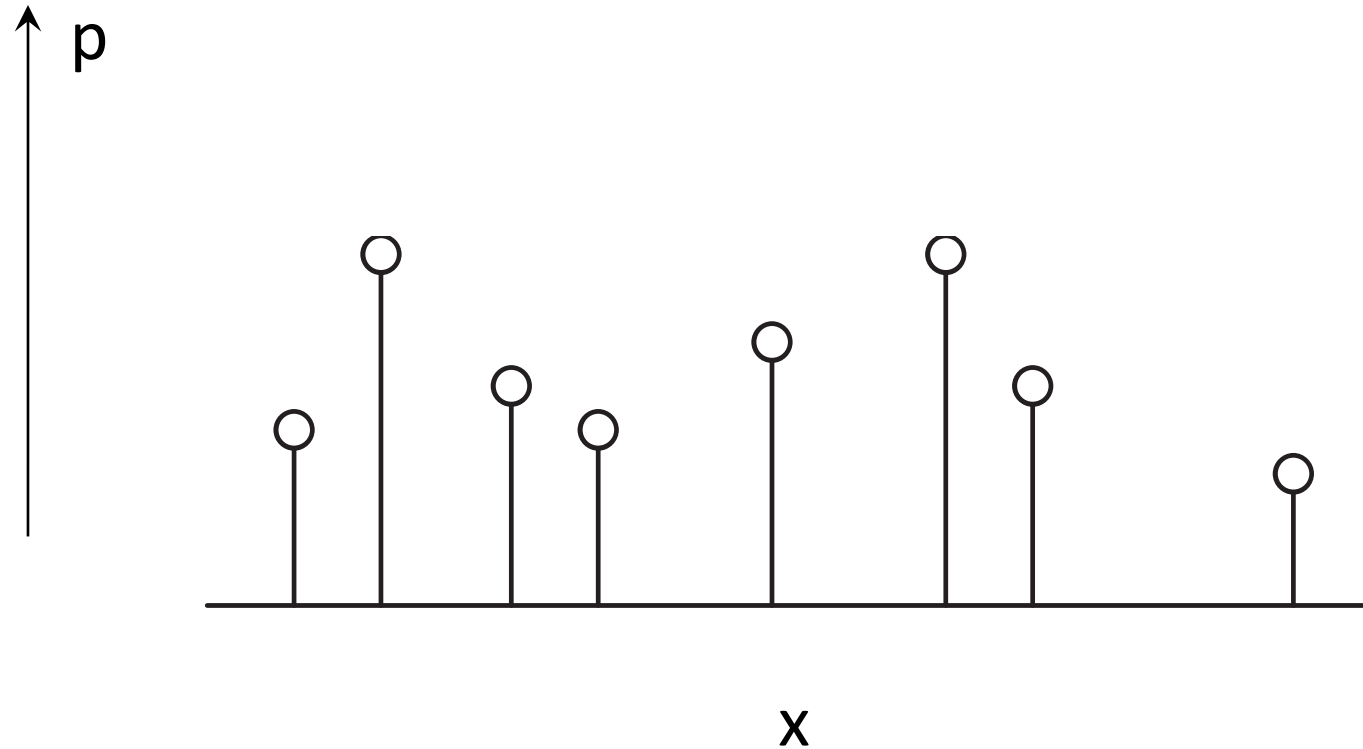
Probability itself is a modeling assumption

- There may be **no true “underlying” distribution** behind the data.
- Even if it exists, we can only observe **a finite sample set** of it.
- Models must **extrapolate** from observations to represent a distribution.
- Leads to **overfitting vs. underfitting** (just as in discriminative models)



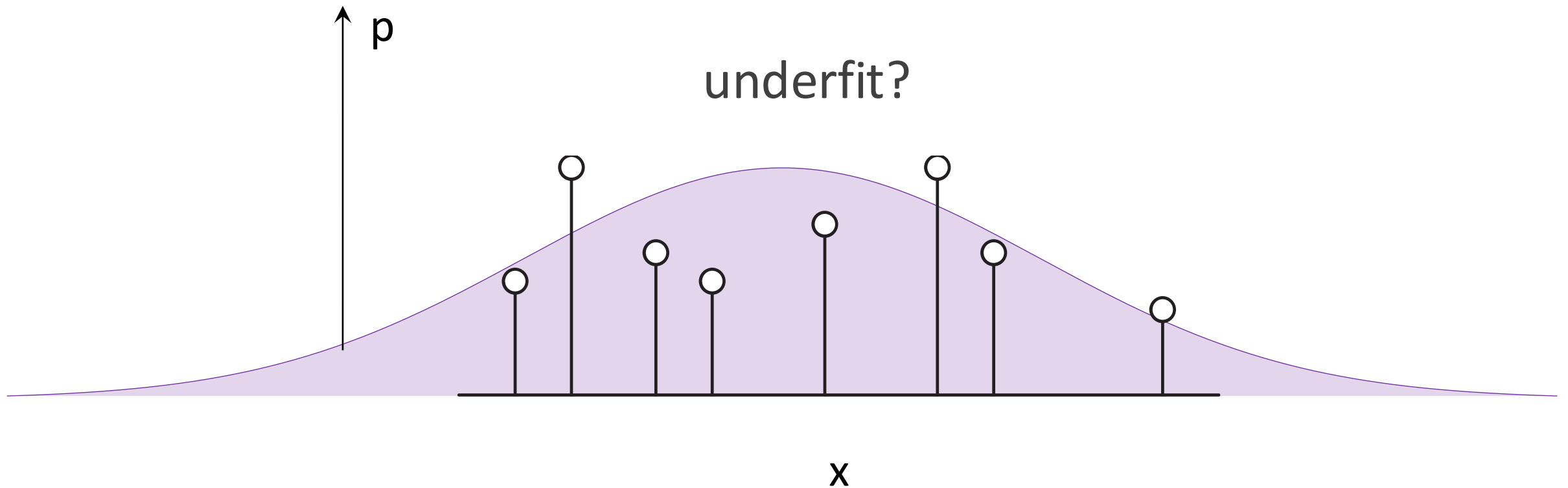
Probability itself is a modeling assumption

- We can only observe a **finite sample set** of an unknown distribution.



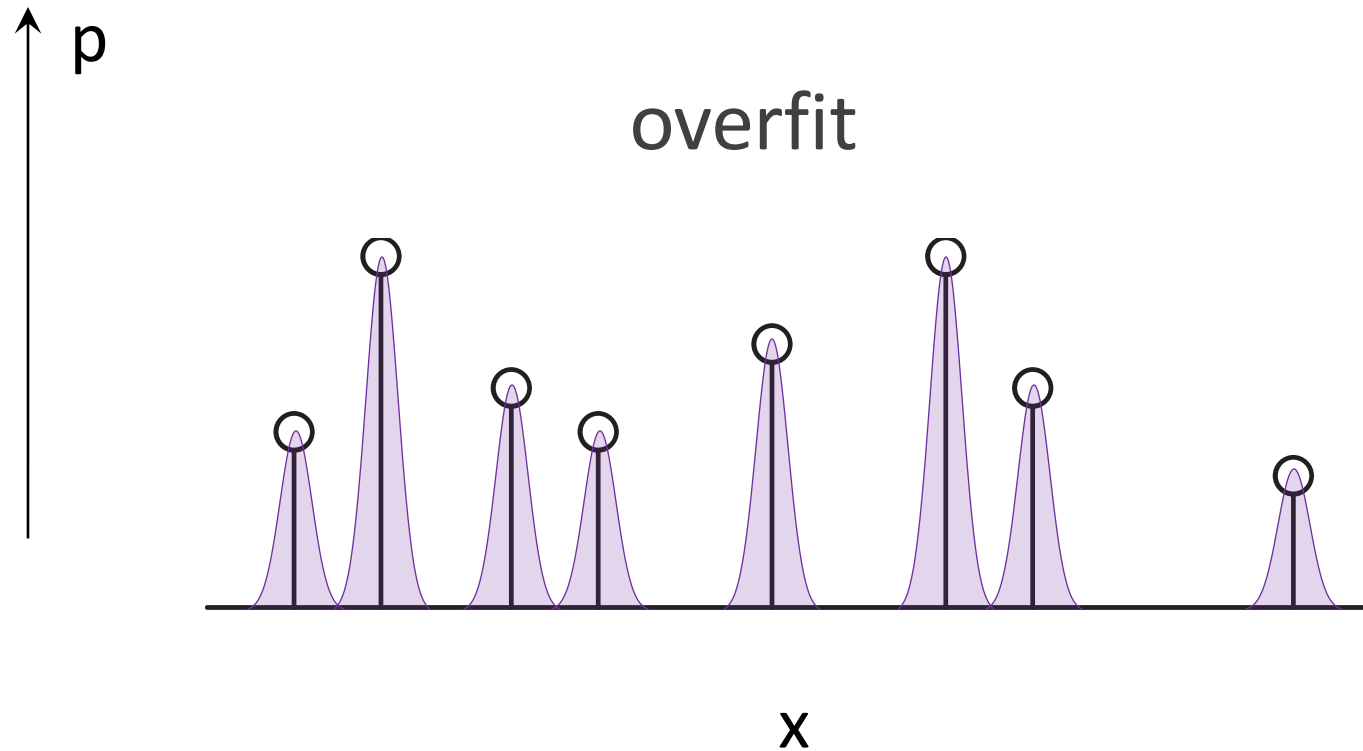
Probability itself is a modeling assumption

- **Underfitting:** the probabilistic model is too simple.



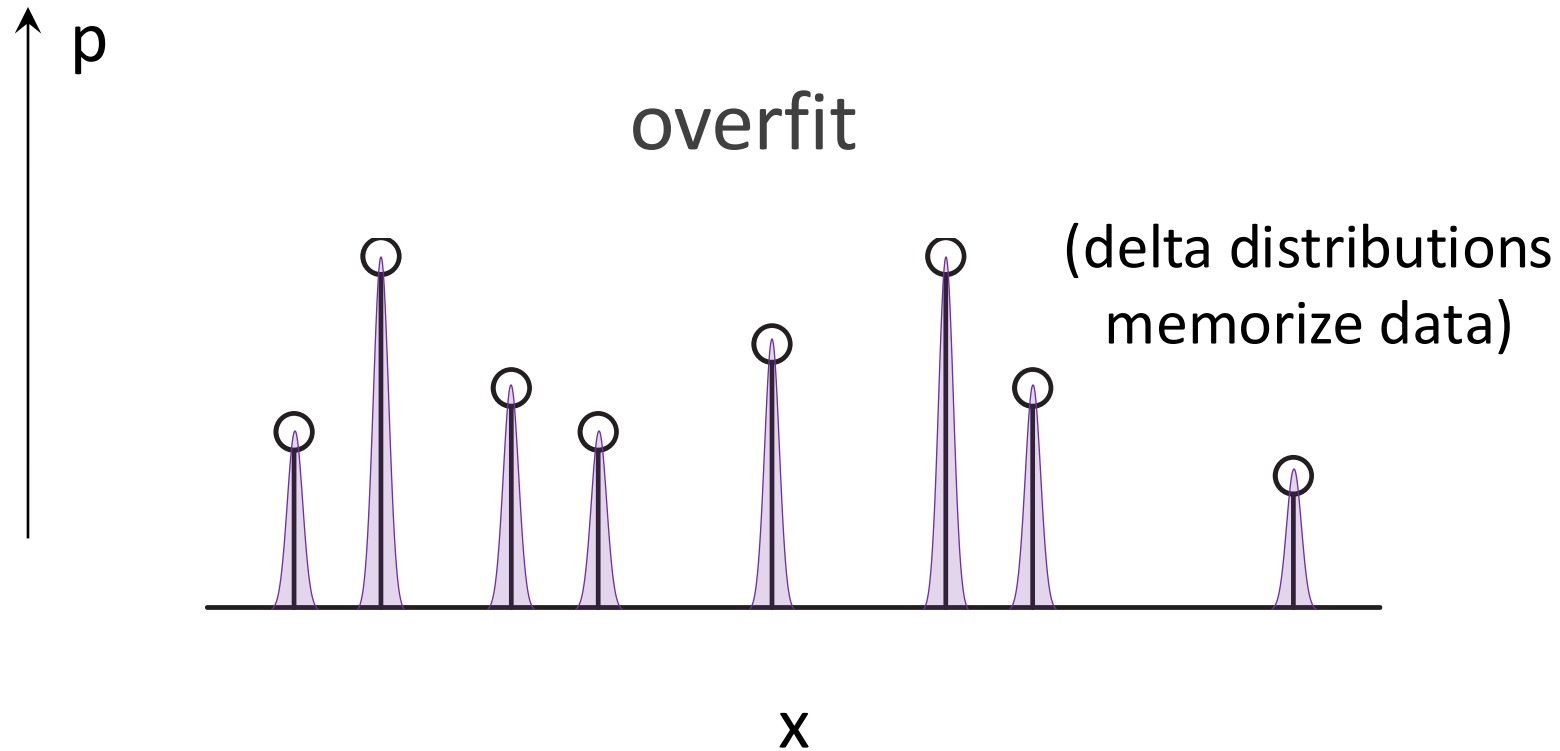
Probability itself is a modeling assumption

- **Overfitting:** the probabilistic model does not extrapolate/generalize



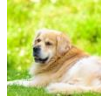
Probability itself is a modeling assumption

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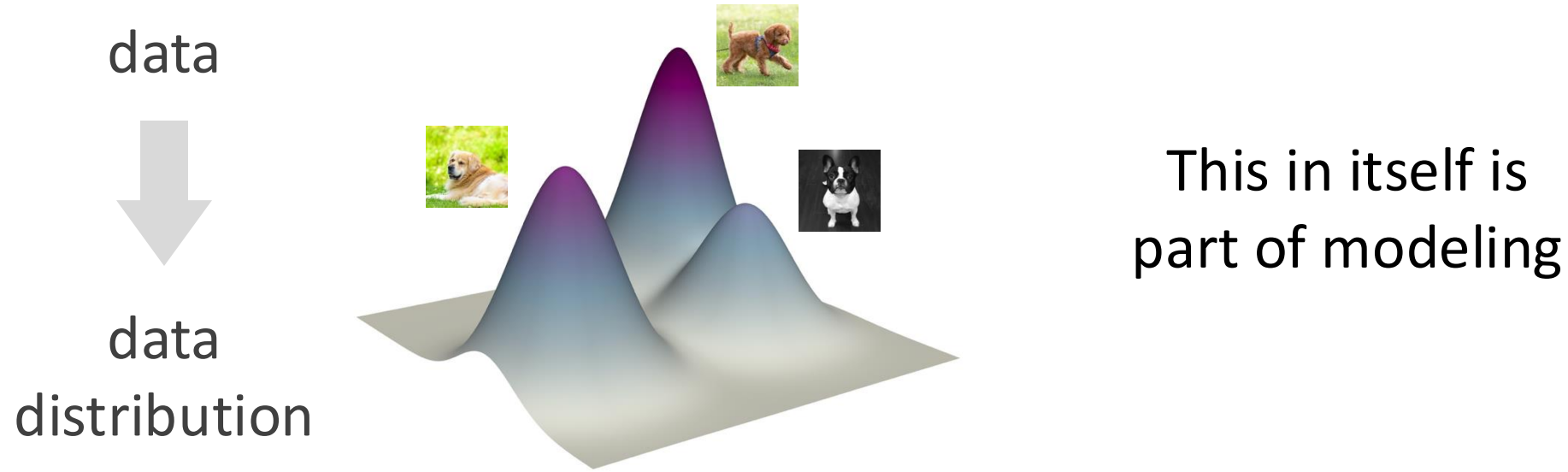


Generative Modeling is Probabilistic Modeling

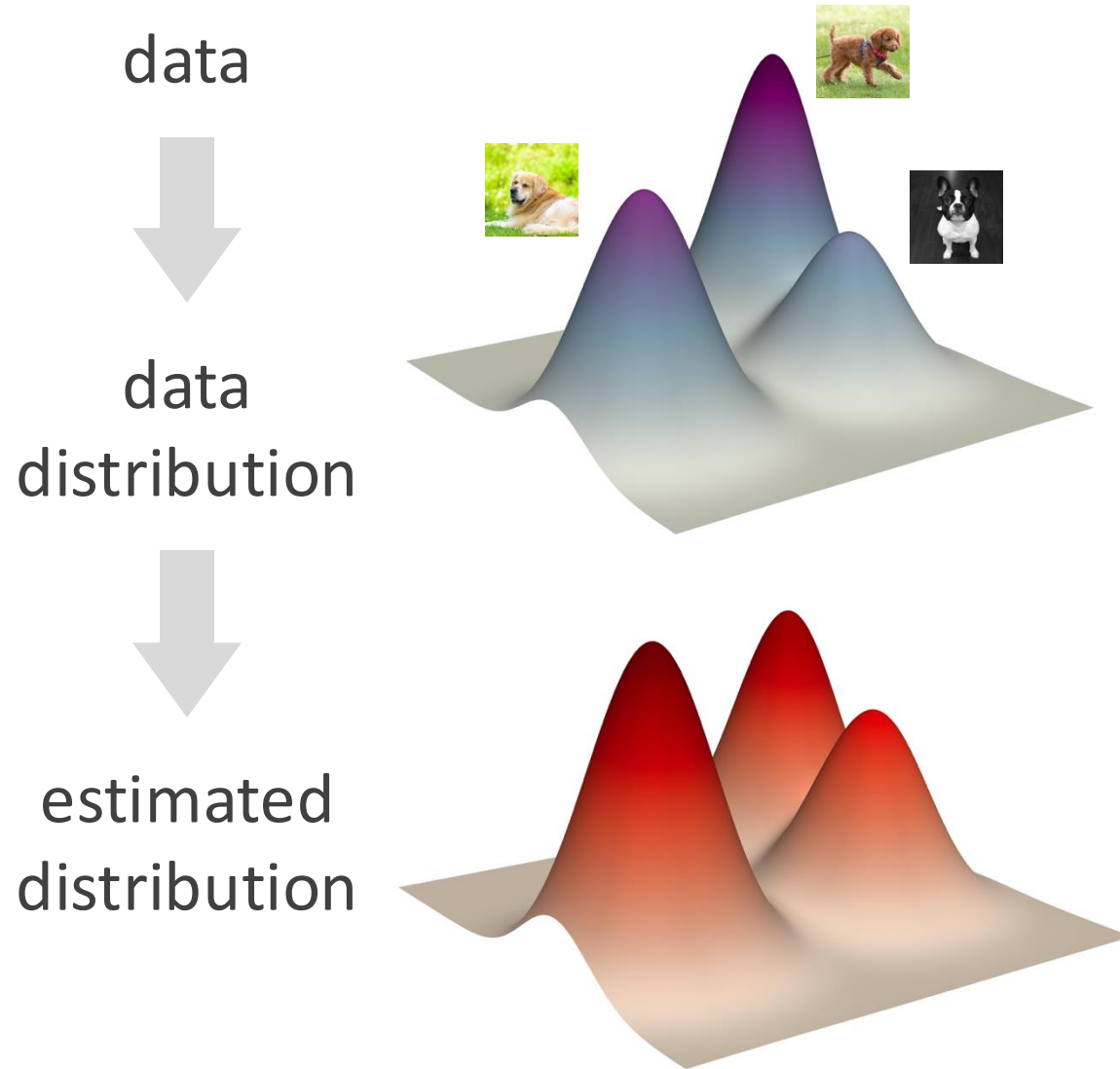
data



Generative Modeling is Probabilistic Modeling



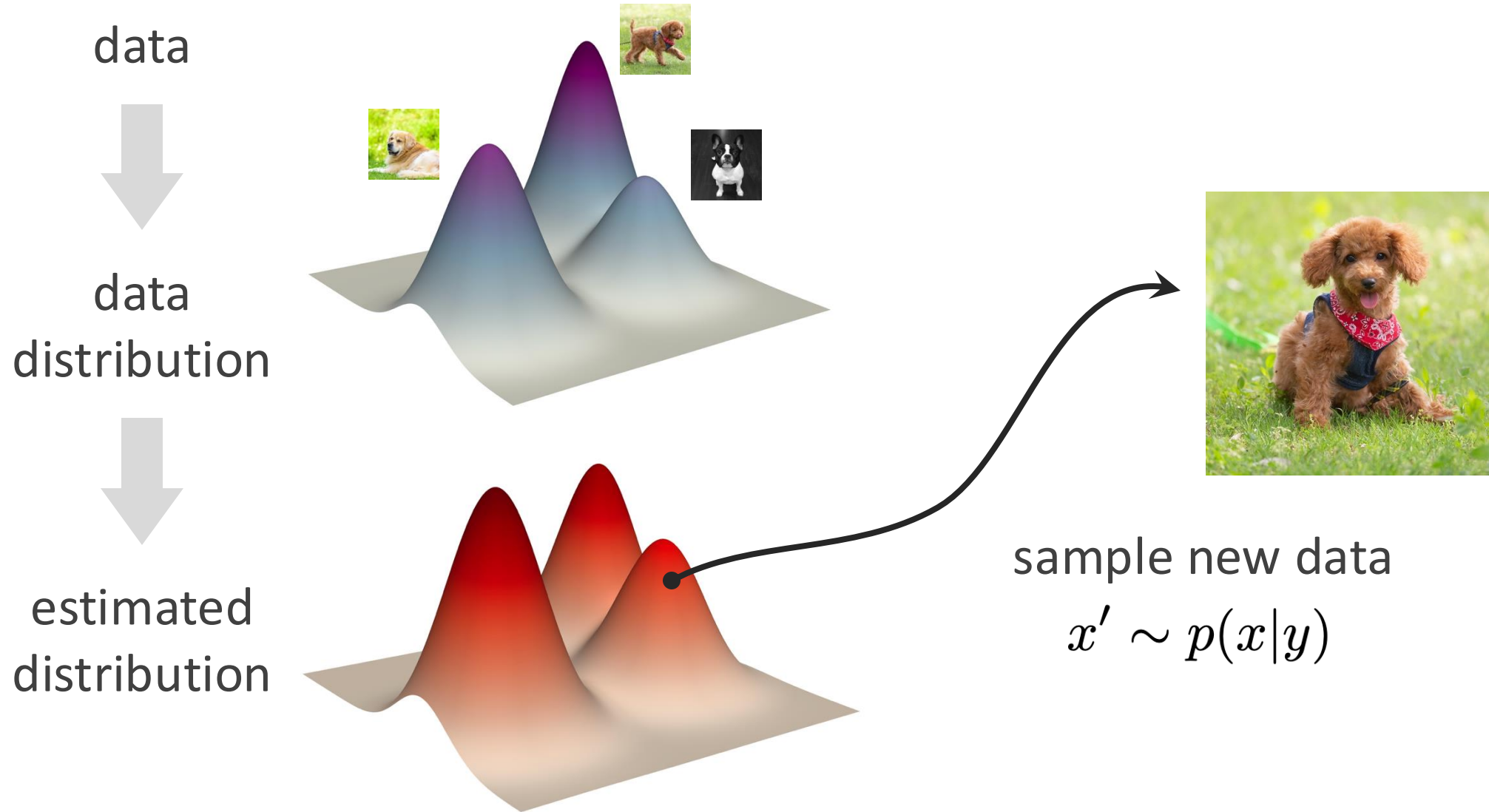
Generative Modeling is Probabilistic Modeling



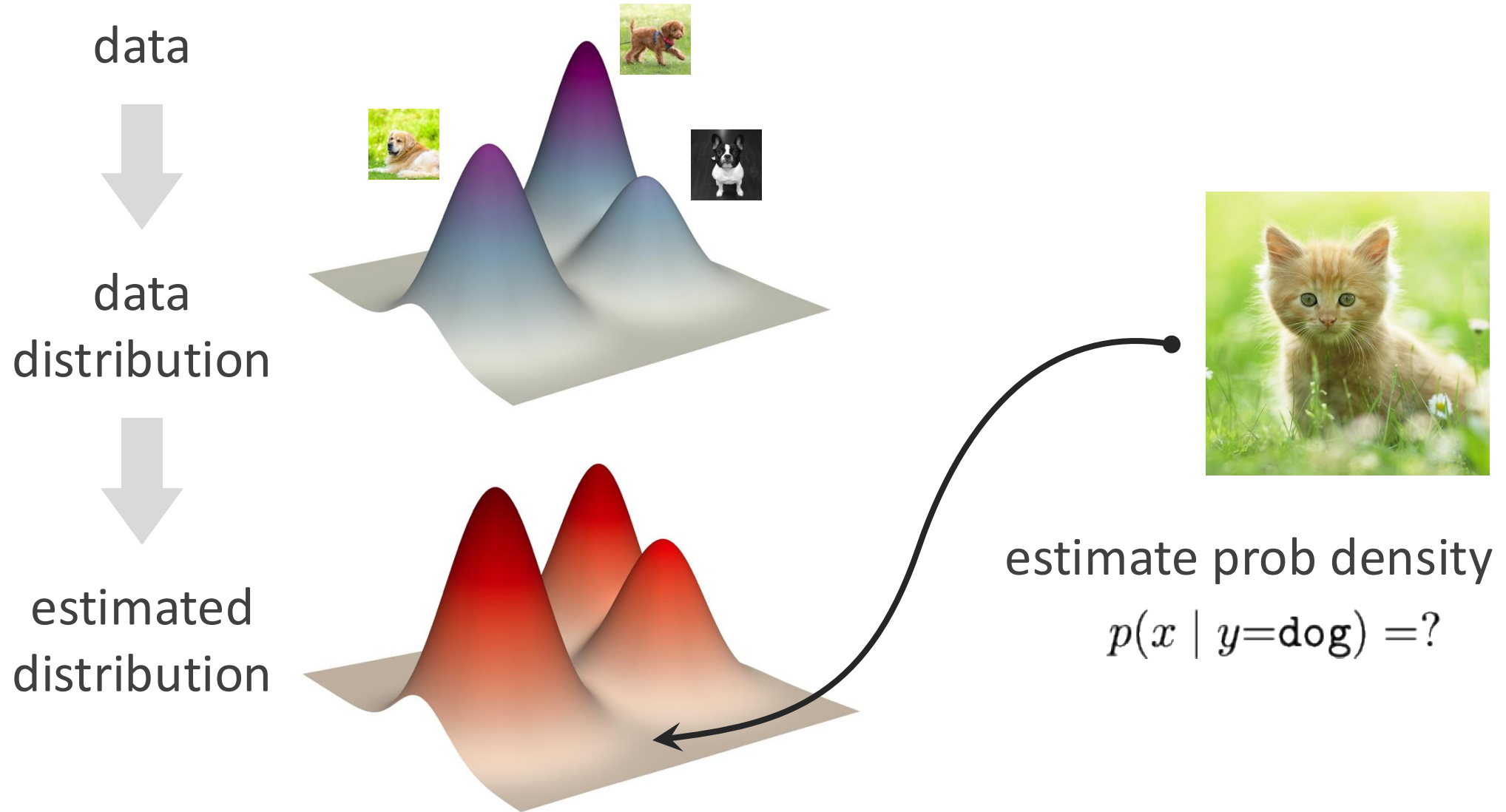
Optimize a loss function

$$\mathcal{L}(\text{data distribution}, \text{estimated distribution})$$

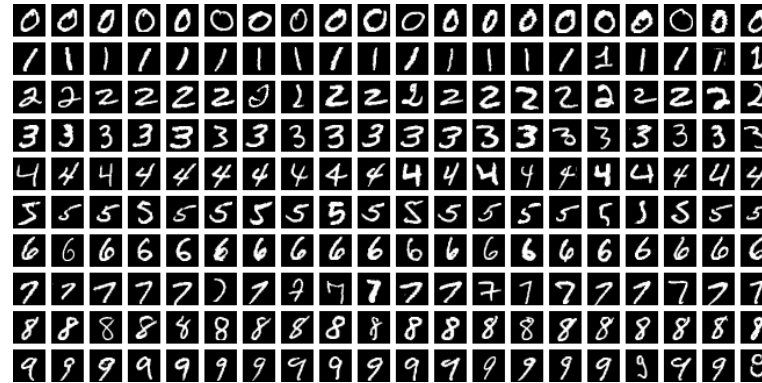
Generative Modeling is Probabilistic Modeling



Generative Modeling is Probabilistic Modeling



Case study: a minimalist generative model?



data set

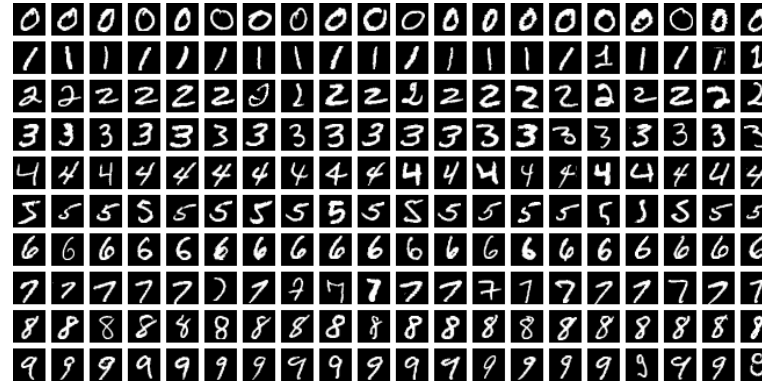


random
index



- Is this a generative model? **Yes**
- Is this a “good” generative model? **No - not generalize?**

Case study: a minimalist generative model?



data set



random
index

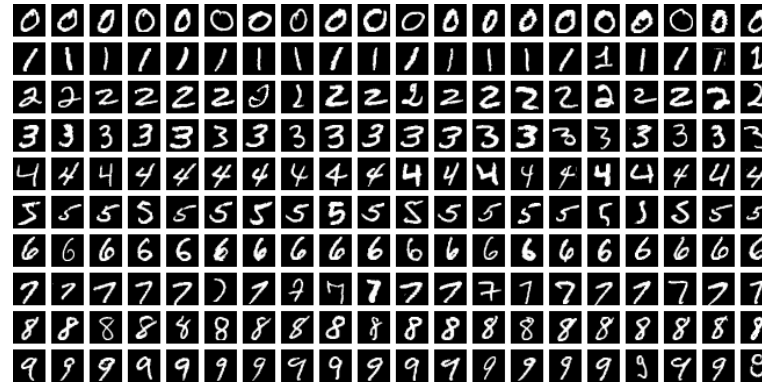


retrieve
+ random noise



- Is this a new sample? **Yes**
- Is this a “good” new sample? **No - not “within distribution”?**

Case study: a minimalist generative model?



data set



2 random
indexes

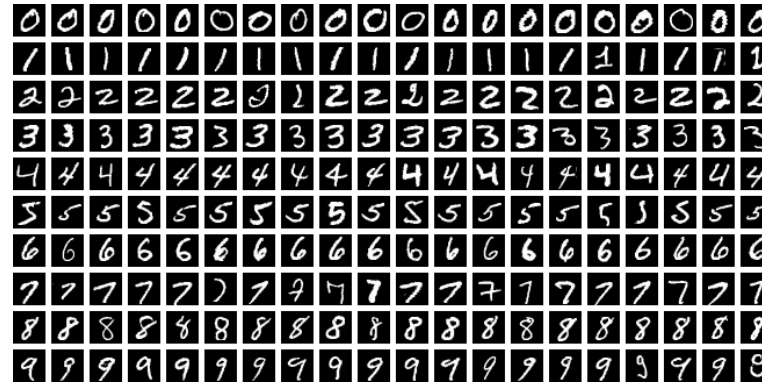


retrieve
and merge



- Is this a new sample? **Yes**
- Is this a “good” new sample? **No - not “within distribution”?**

Case study: a minimalist generative model?



data set



random
index



retrieve
and **colorize**



- Is this a new sample? **Yes**
- Is this a “good” new sample? **No? It depends.**

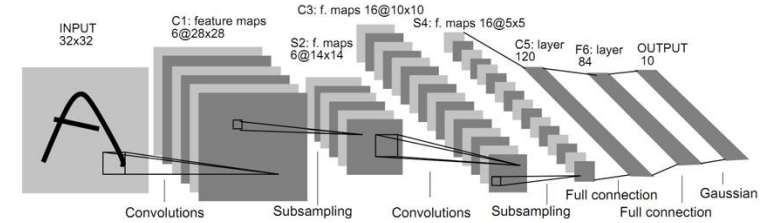
Generative Modeling is Probabilistic Modeling

Notes:

- Generative models involve probabilistic models hypothesized by humans.
- Probabilistic modeling is not just the work of neural nets (even in the context of deep learning).
- Probabilistic modeling is a popular way, but not the only way.
- *"All models are wrong, but some are useful."**

Deep Generative Models

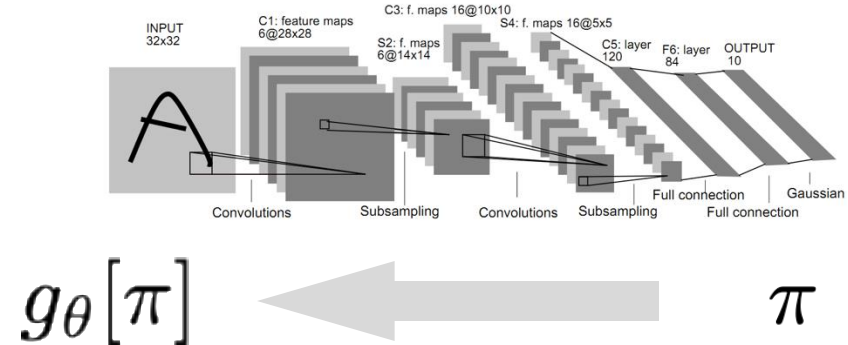
- Deep learning is **representation learning**
- **Discriminative:** represent data **instances**
 - one data point: $x \rightarrow f_{\theta}(x)$
 - loss w.r.t. one label: $\mathcal{L}(y, f_{\theta}(x))$



Deep Generative Models

- Deep learning is **representation learning**
- **Discriminative**: represent data **instances**

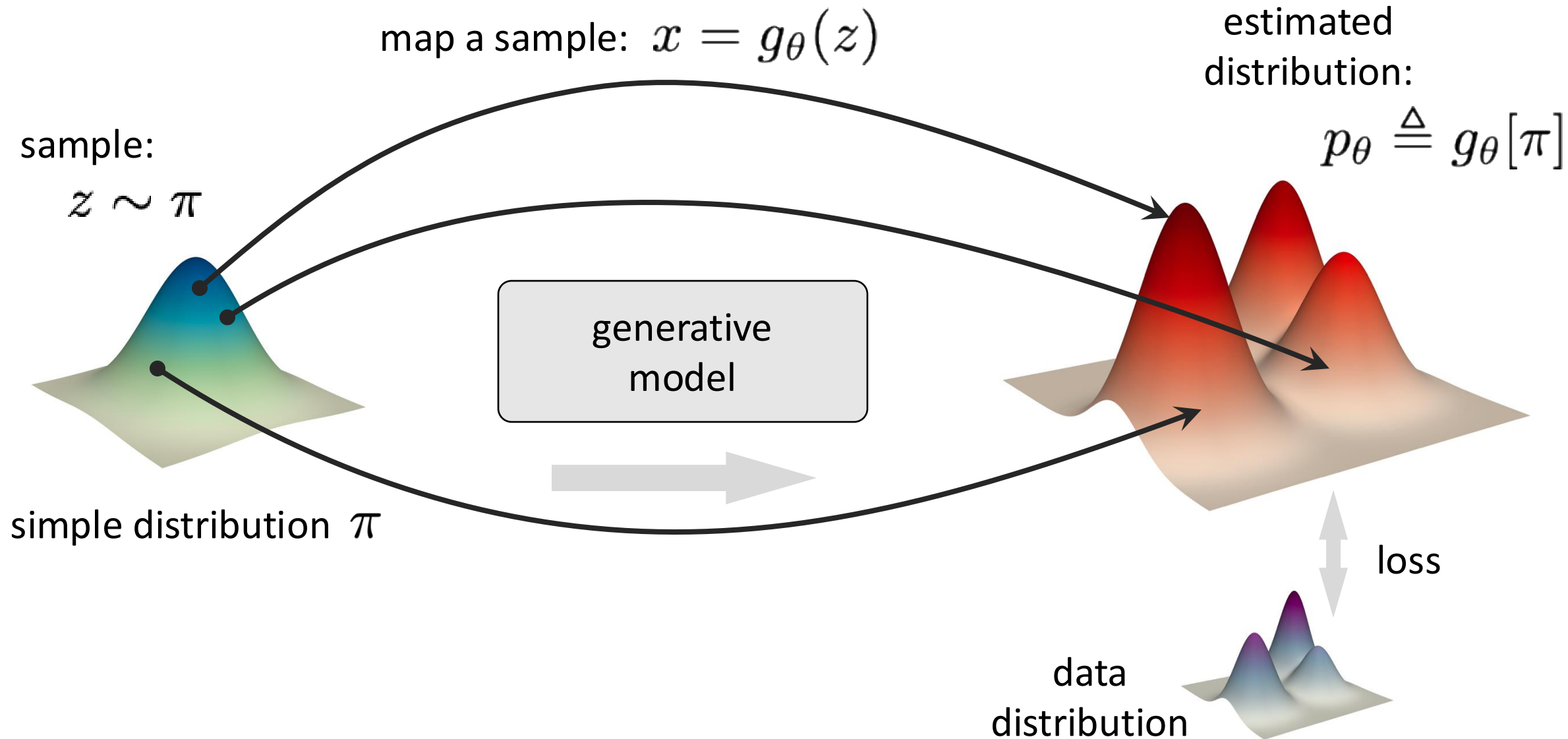
- one data point: $x \rightarrow f_{\theta}(x)$
- loss w.r.t. one label: $\mathcal{L}(y, f_{\theta}(x))$



- **Generative**: represent data **distribution**

- distribution-to-distribution mapping: $\pi \rightarrow g_{\theta}[\pi]$ (example: π is Gaussian)
- loss w.r.t. true (but known) distribution: $\mathcal{L}(p_{\text{data}}, g_{\theta}[\pi])$

Learning to Represent Distributions



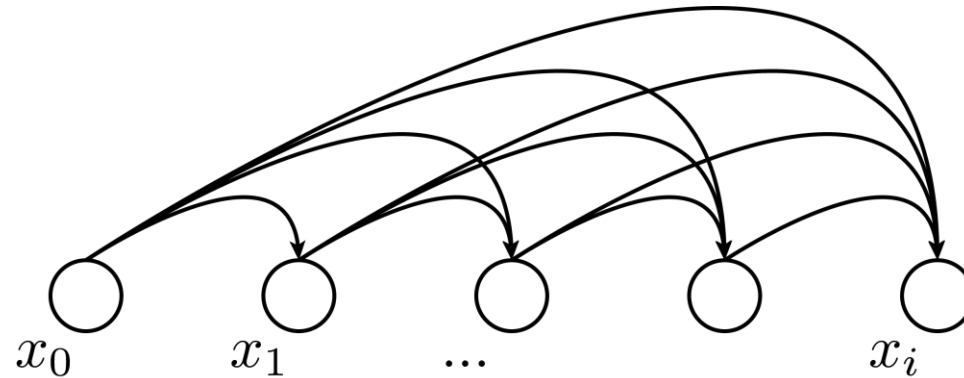
Learning to Represent Distributions

- Not all parts of a generative model are learned.

Case study:

Autoregressive model

This dependency structure is designed (not learned).



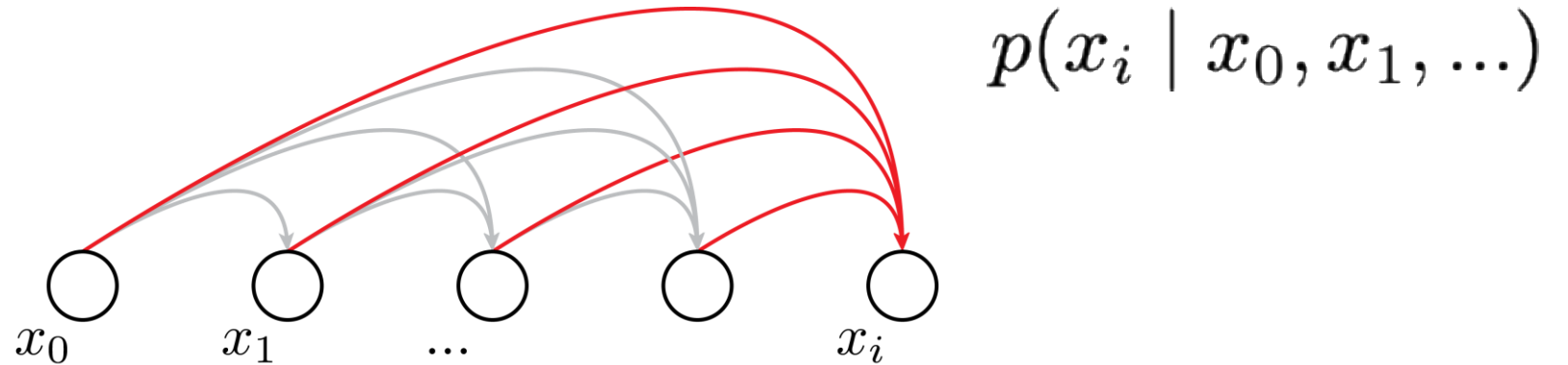
Learning to Represent Distributions

- Not all parts of a generative model are learned.

Case study:

Autoregressive model

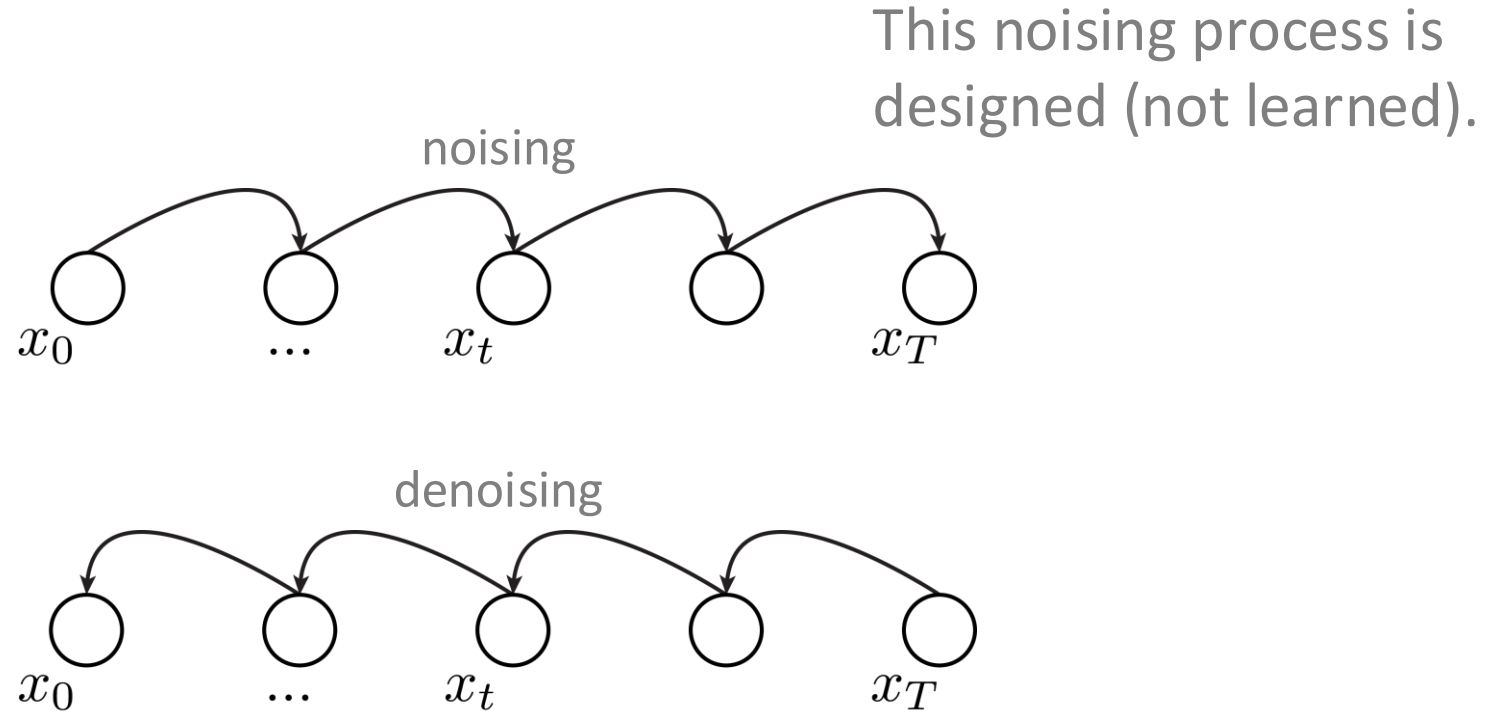
This probability is learned



Learning to Represent Distributions

- Not all parts of a generative model are learned.

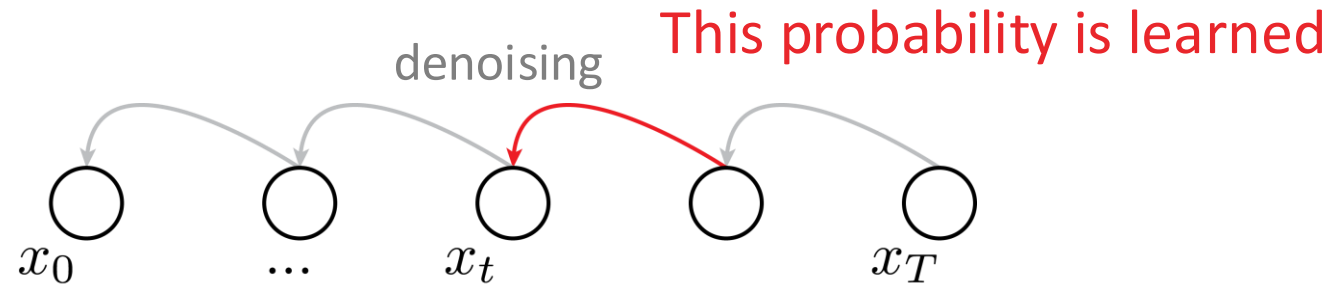
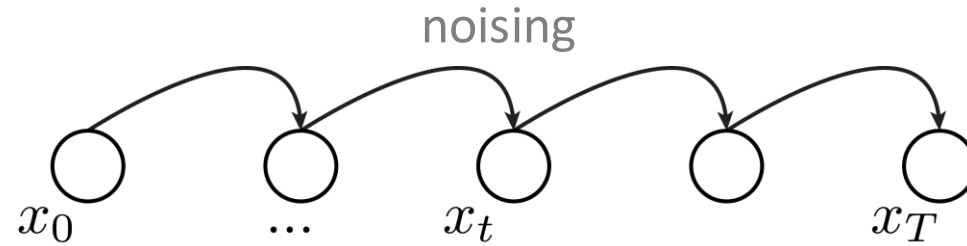
Case study:
Diffusion model



Learning to Represent Distributions

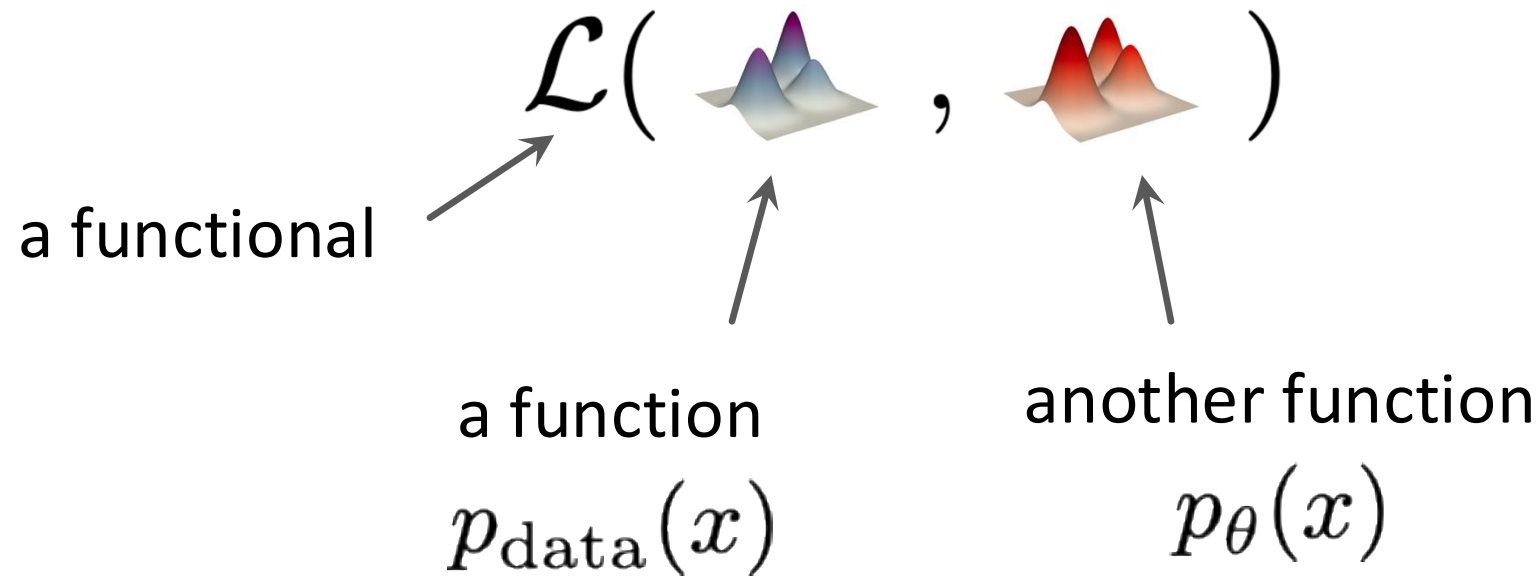
- Not all parts of a generative model are learned.

Case study:
Diffusion model



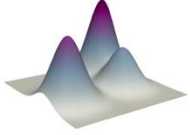
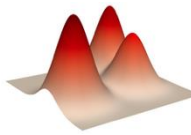
Objective Functions in Generative Models

- The objective is a *functional*: it operates on functions



Objective Functions in Generative Models

- Case study: Kullback–Leibler (KL) Divergence

$$\mathcal{L}(\text{  , \text{ )$$

x is a dummy
variable

$$\mathcal{D}_{\text{KL}}(p_{\text{data}} \parallel p_{\theta}) \triangleq \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(x)}{p_{\theta}(x)} \right]$$

$\mathcal{D}$'s inputs: $p_{\theta}, p_{\text{data}}$

Objective Functions in Generative Models

- Likelihood: $-\mathbb{E}_{p_{\text{data}}}[\log p_{\theta}(x)]$ MLE, VAE, Normalizing Flows
- Divergence: $\mathcal{D}_{\text{KL}}(p_{\text{data}} \parallel p_{\theta})$ (same as above)
- Adversarial: $\mathcal{D}_f(p_{\text{data}}, p_{\theta})$ GAN
- Score-based: $\mathbb{E}_{p_{\text{data}}} \|\nabla_x \log p_{\text{data}} - \nabla_x \log p_{\theta}\|^2$ Diffusion, Energy-based, ...

What makes the objectives difficult to design?

- Analytical formulations
 - Only for limited families (Gaussian, uniform, categorical, ...)
 - Low-dimensional
 - Example: Gaussian $\Rightarrow$ L2 loss
- Monte Carlo estimation
 - Distributions unknown in analytical form
 - We can only sample from them
 - Example: $x, y \sim p_{\text{data}}(x, y)$
- Divergence measures
 - often not directly computable (a surrogate is used)
 - Example: GAN

Elements of Deep Generative Models

- **Formulation:** a real-world problem => probabilistic modeling
 - designed structures
 - learnable components
- **Representation:** use deep neural nets to represent distributions
- **Objectives:** discrepancy between distributions
- **Optimization:** differentiable, computable, and tractable
- **Inference:**
 - sampler: able to produce new samples
 - density estimator: (optional) able to evaluate $p(x)$

Generative Modeling of Real-world Problems

Generative Modeling of Real-world Problems

- Generative models are about $p(x|y)$

What can be y ?

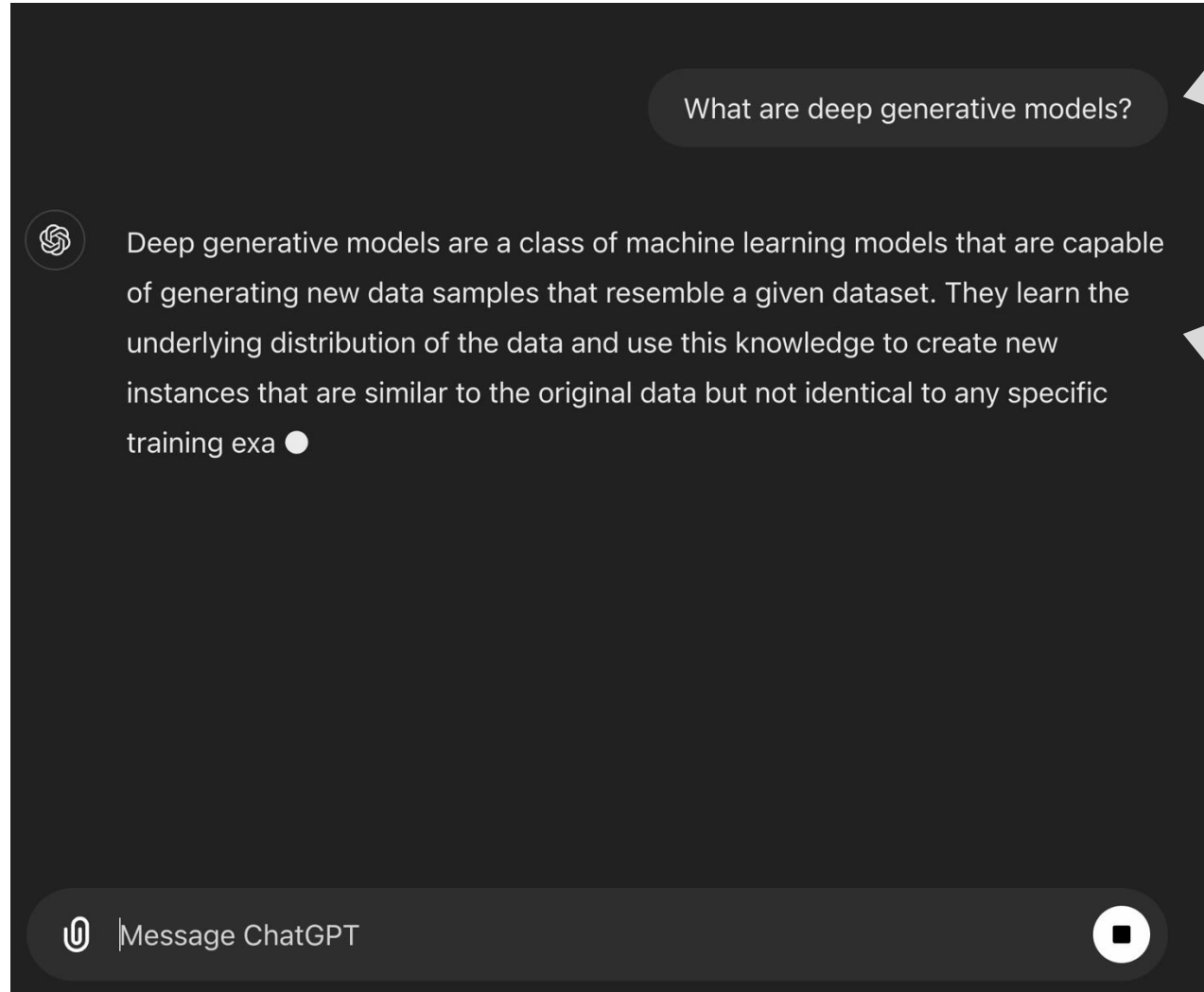
- condition
- constraint
- labels
- attributes
- more abstract
- less informative

What can be x ?

- “data”
- samples
- observations
- measurements
- more concrete
- more informative

Generative Modeling as $p(x|y)$: Case Study

- Natural language conversation



y: prompt

x: response of the chatbot

Generative Modeling as $p(x | y)$: Case Study

- Text-to-image/video generation

*Prompt: teddy bear teaching a course, with
"generative models" written on blackboard*

← y: text prompt



← x: generated visual content

Generative Modeling as $p(x|y)$: Case Study

- Text-to-3D structure generation



“motorcycle”



“mech suit”



“ghost lantern”



“furry fox head”



“dresser”



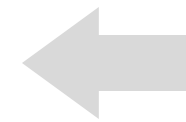
“swivel chair”



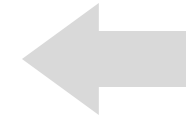
“astronaut”



“mushroom house”



x: generated 3D structures

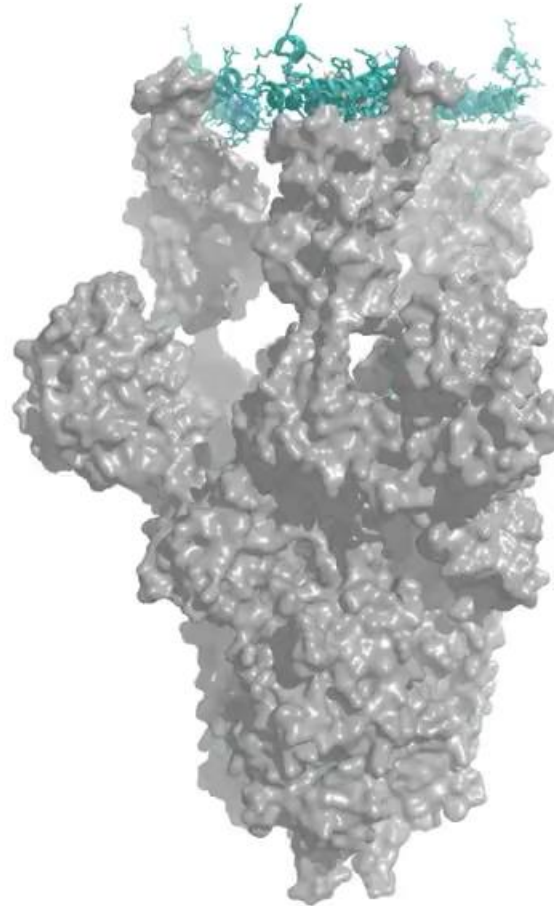


y: text prompt

Generative Modeling as $p(x|y)$: Case Study

- Protein structure generation

y: condition/constraint
(e.g., symmetry)

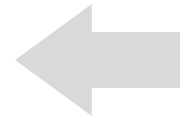


x: generated
protein structures

Generative Modeling as $p(x|y)$: Case Study

- Class-conditional image generation

“red fox”



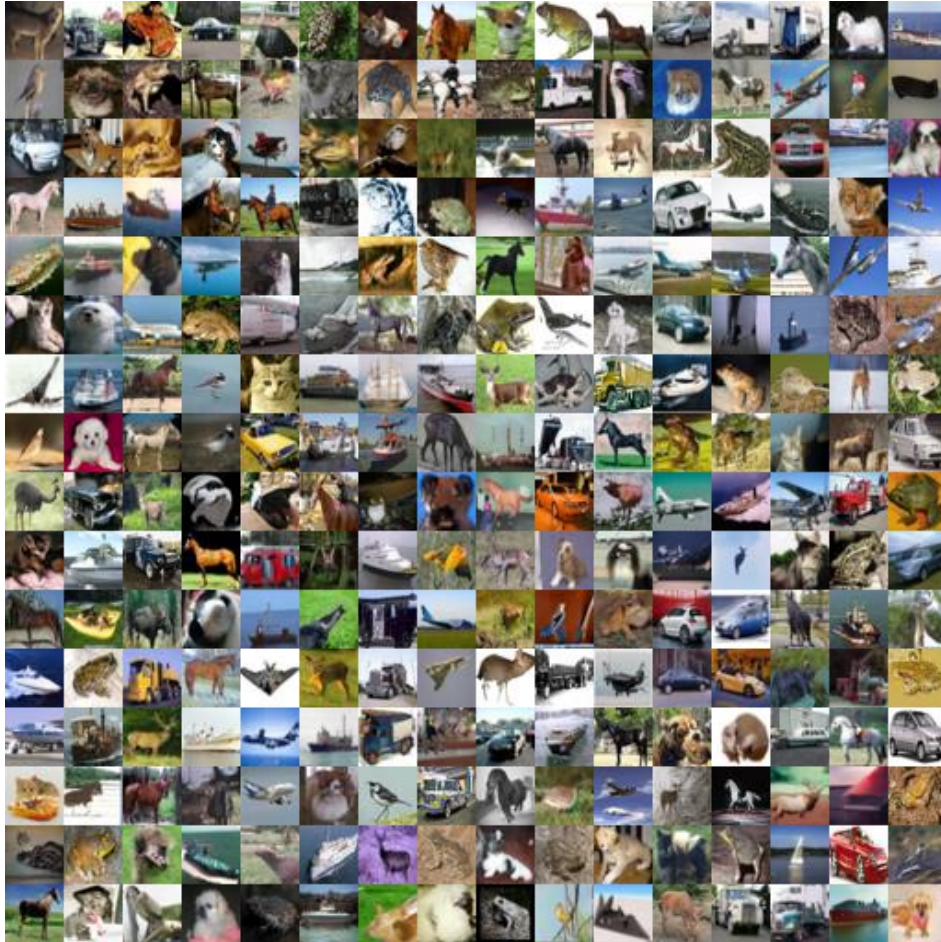
y: class label



x: generated image

Generative Modeling as $p(x|y)$: Case Study

- “Unconditional” image generation



y : an implicit condition

“images following CIFAR10 distribution”

x : generated CIFAR10-like images

- $p(x|y)$: images $\sim$ CIFAR10
- $p(x)$: all images

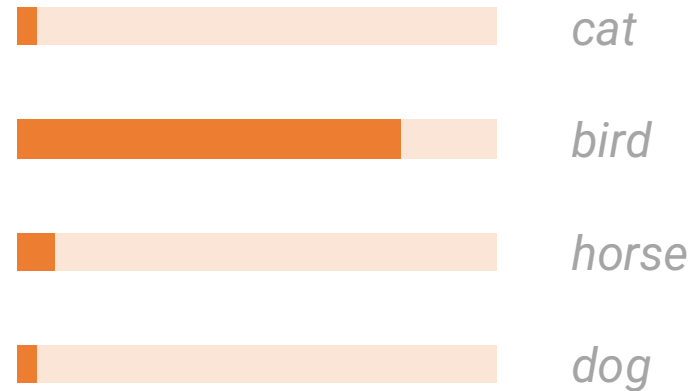
Generative Modeling as $p(x|y)$: Case Study

- **Classification** (often not viewed as generative)

y: an image as the “condition”



x: probability of classes
conditioned on the image



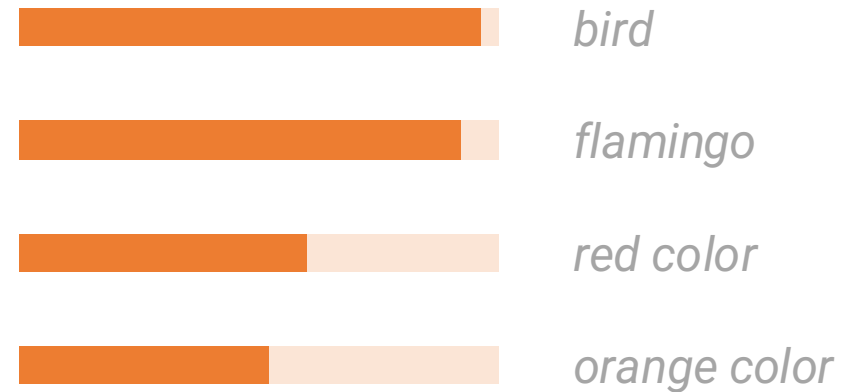
Generative Modeling as $p(x|y)$: Case Study

- Open-vocabulary recognition

y: an image as the “condition”



x: plausible descriptions
conditioned on the image



.....

...

Generative Modeling as $p(x | y)$: Case Study

- Image captioning

y : an image as the “condition”



x : plausible descriptions
conditioned on the image

a baseball player with a catcher and umpire on top of a baseball field.
a baseball player is sliding into a base.
a baseball player swings at a pitch with the pitcher and umpire behind him.
baseball player with bat in the baseball game.
a batter in the process on the bat in a baseball game.

Generative Modeling as $p(x|y)$: Case Study

- Visual dialogue

User What is unusual about this image?



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

y: image and text prompt

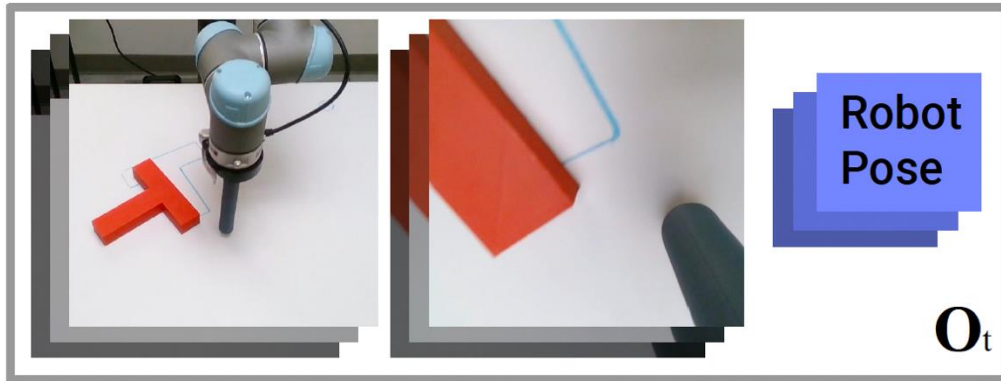
GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

x: response of the chatbot

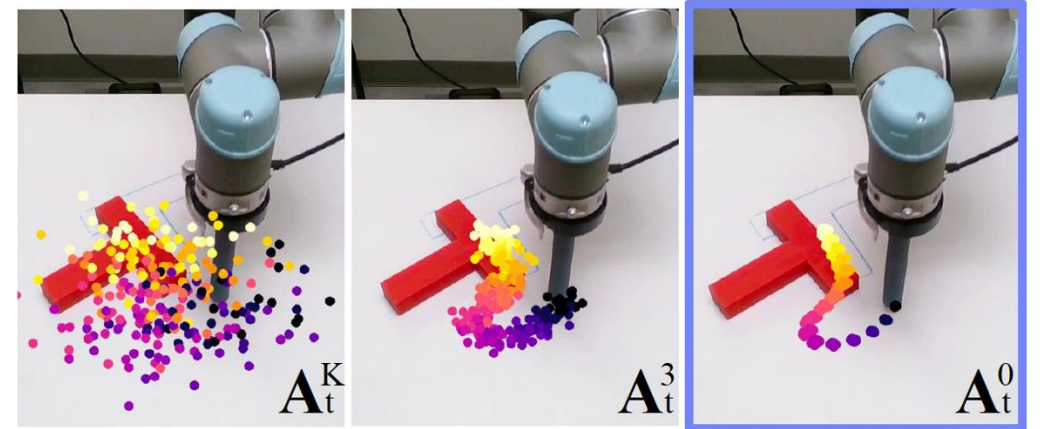
Generative Modeling as $p(x|y)$: Case Study

- Policy Learning in Robotics

y : visual and other
sensory observations



x : policies
(probability of actions)



Generative Modeling of Real-world Problems

- Generative models are about $p(x|y)$

What can be y ?

- condition
- constraint
- labels
- attributes
- more abstract
- less informative

What can be x ?

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- samples
- observations
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Generative Modeling of Real-world Problems

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What can be y ?

- ~~condition~~
- ~~constraint~~
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- ~~more abstract~~
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What can be x ?

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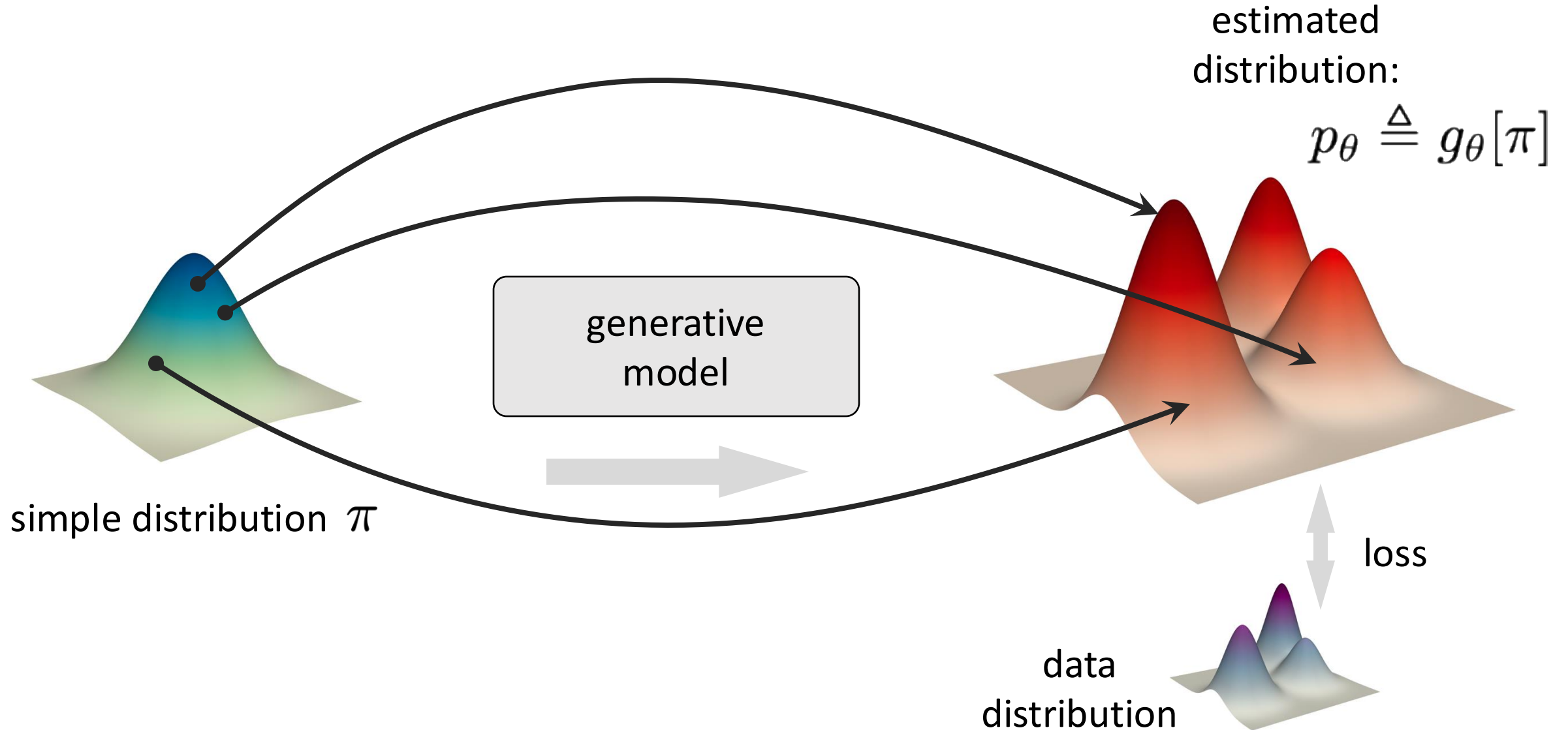
Generative Modeling of Real-world Problems

- Generative models are about $p(x|y)$

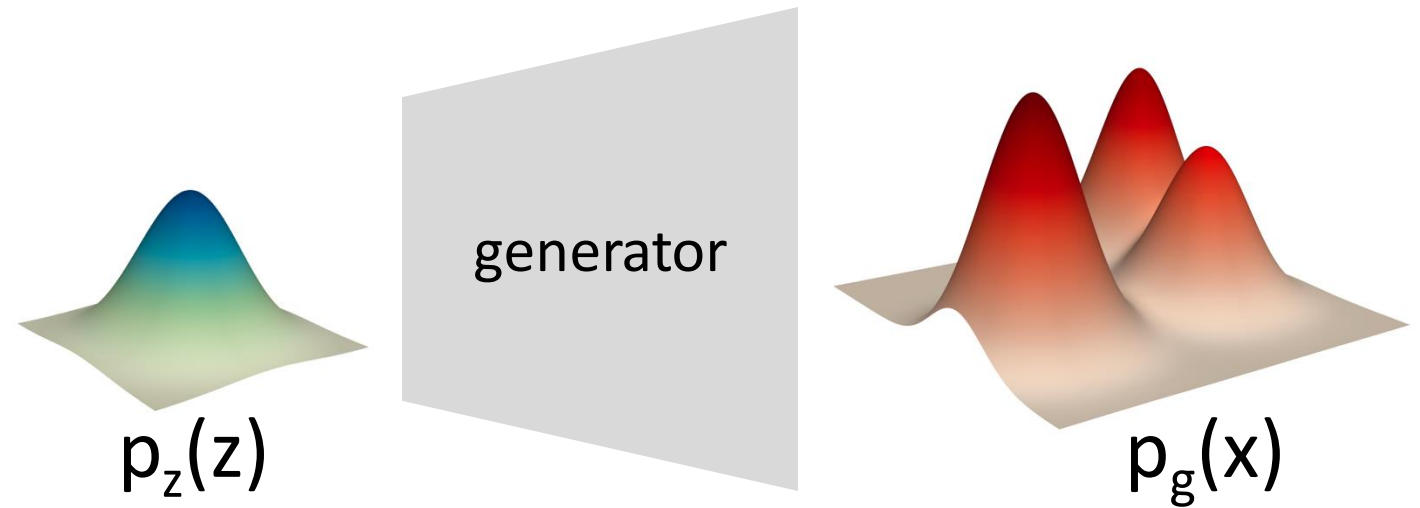
- There is no conceptual obstacle to formulating any real-world problem from a generative perspective.
- Generative modeling is a way of problem-solving.

Families of Generative Models

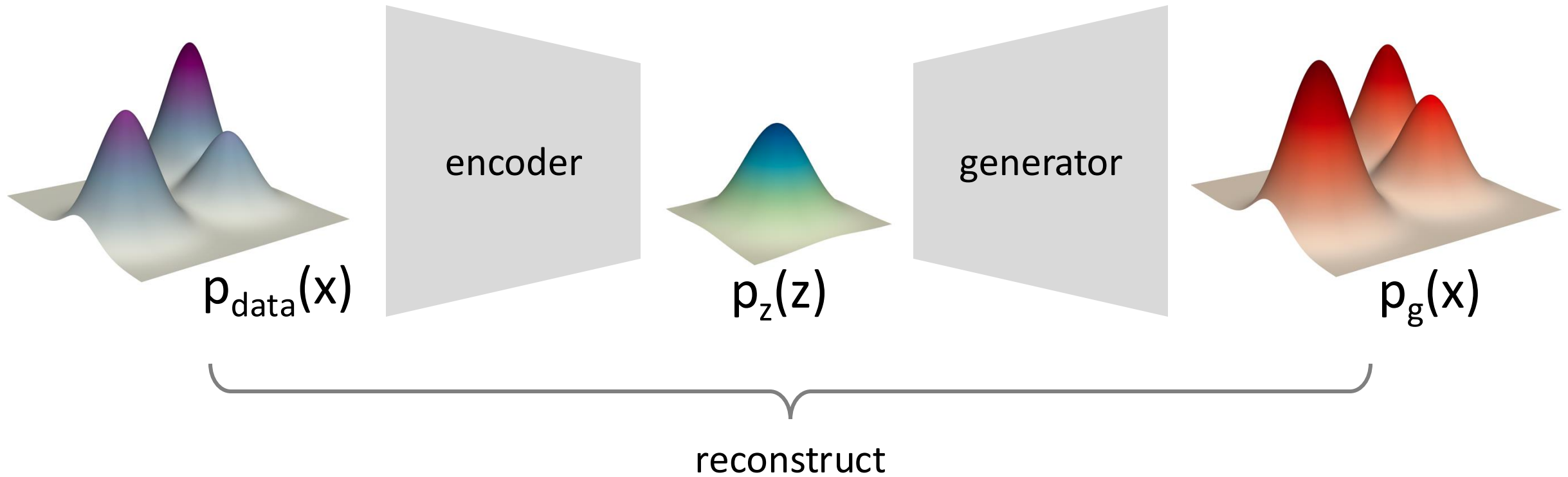
[Recap] Learning to Represent Distributions



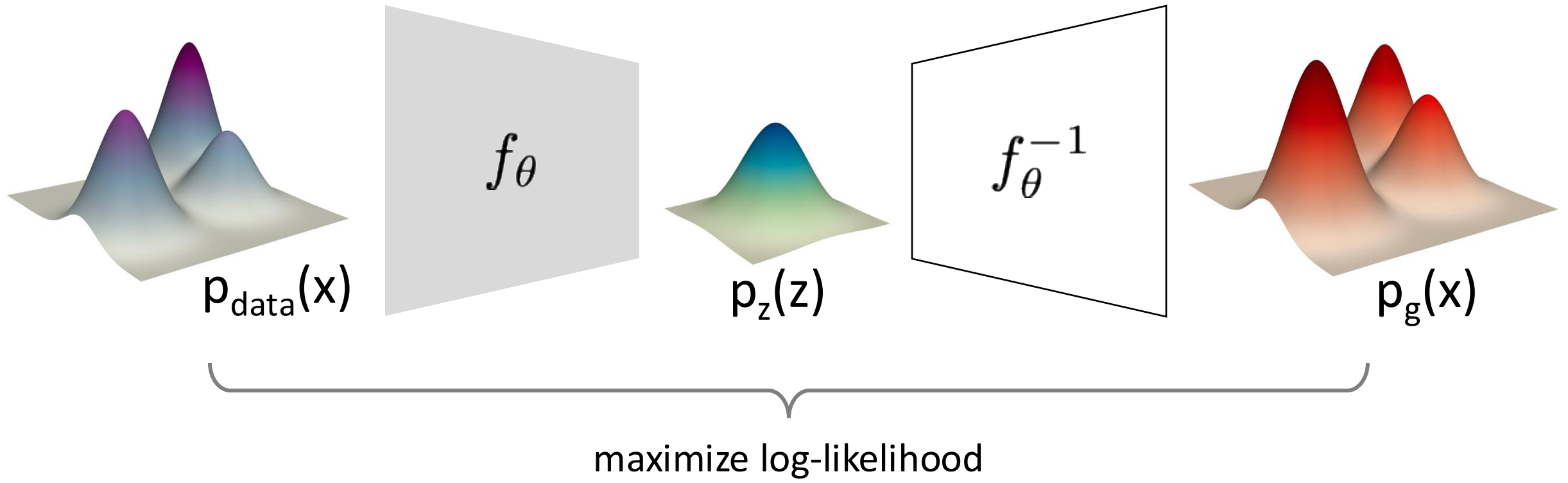
[Recap] Learning to Represent Distributions



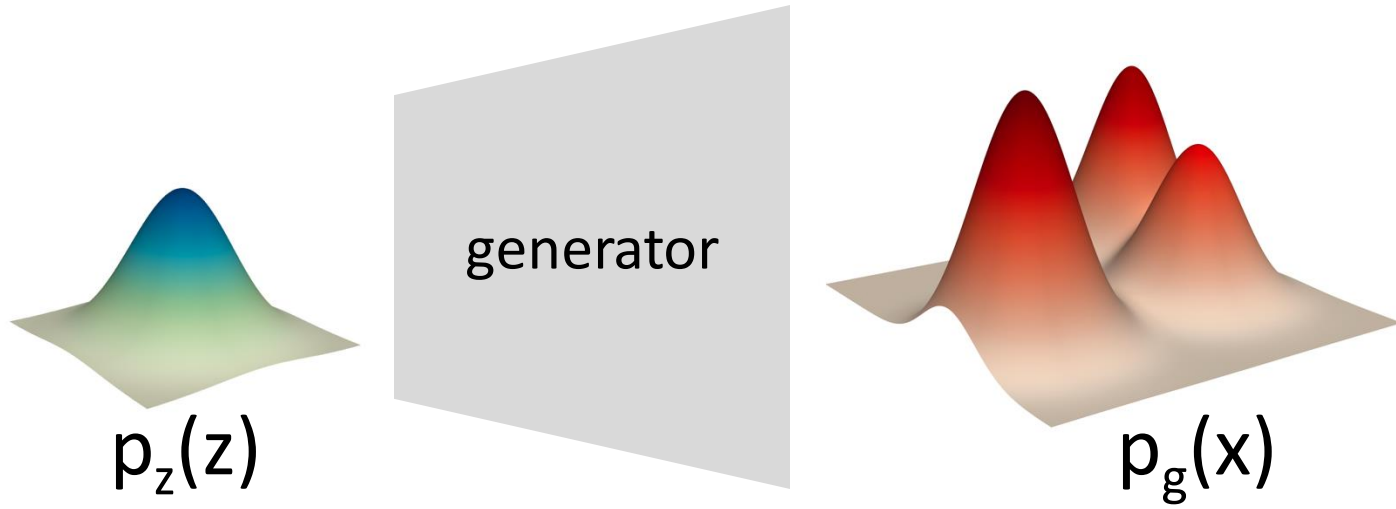
Variational Autoencoder (VAE)



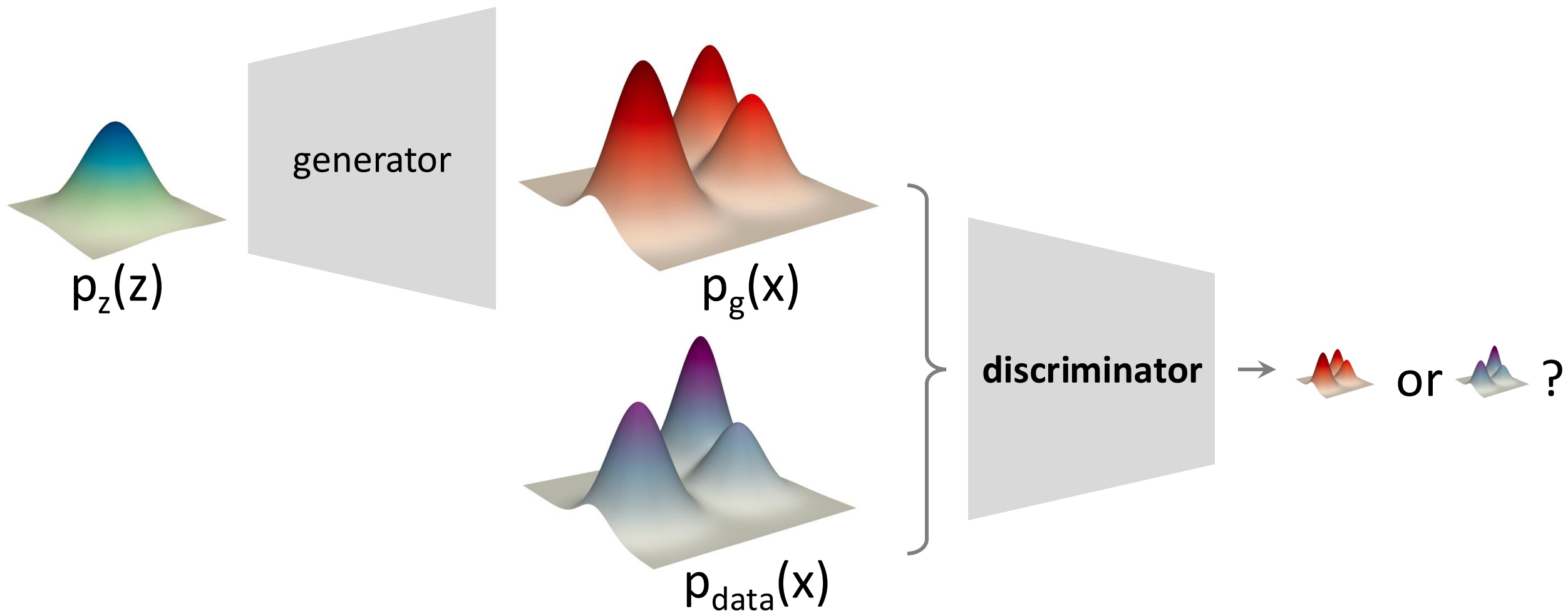
Normalizing Flows



Generative Adversarial Net (GAN)

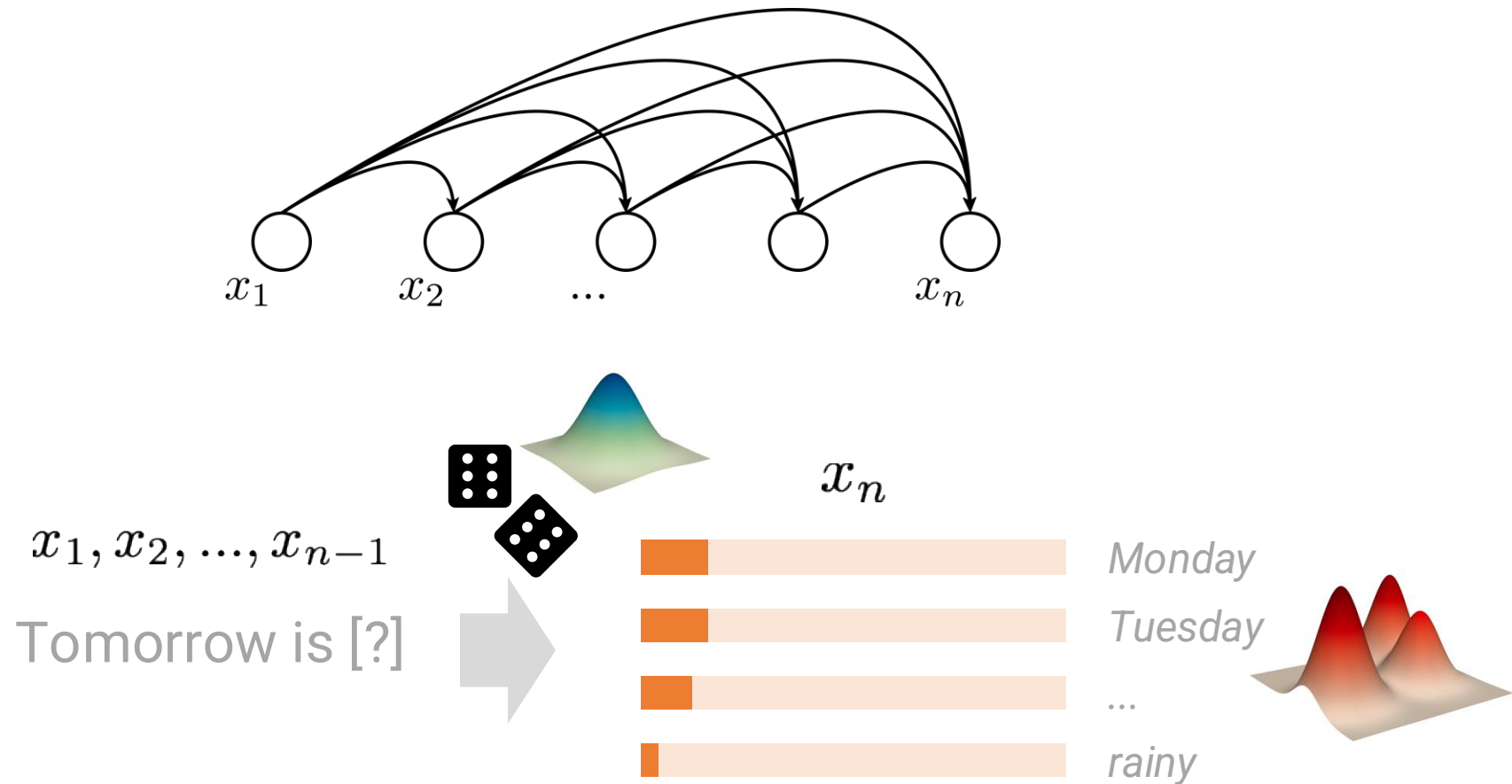


Generative Adversarial Net (GAN)

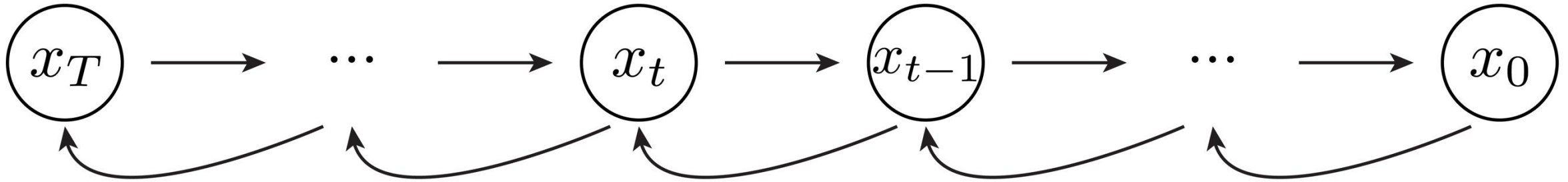
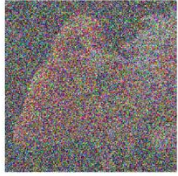
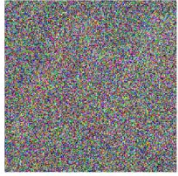
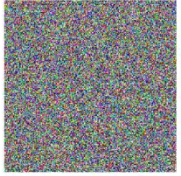


Autoregressive Models (AR)

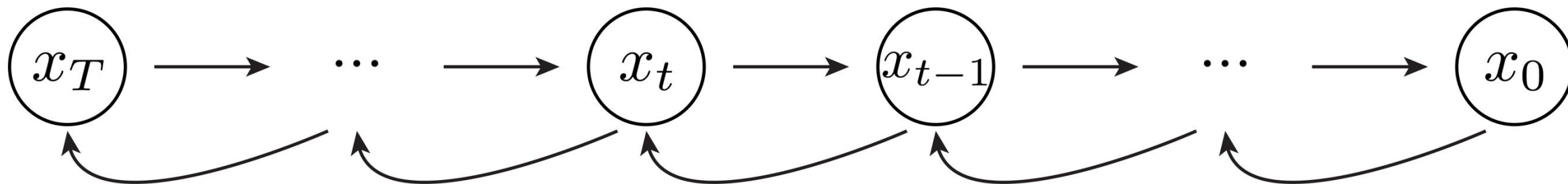
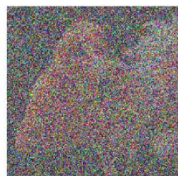
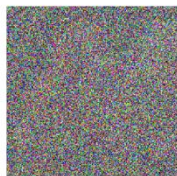
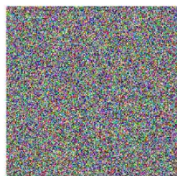
$$p(x_1, x_2, \dots, x_n) = p(x_1)p(x_2 \mid x_1) \dots p(x_n \mid x_1, x_2, \dots, x_{n-1})$$



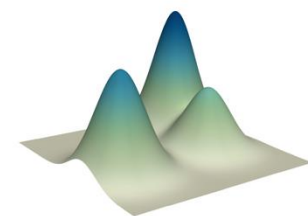
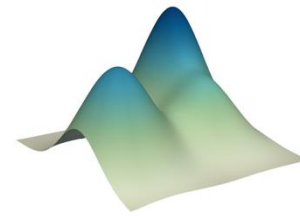
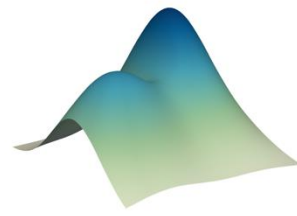
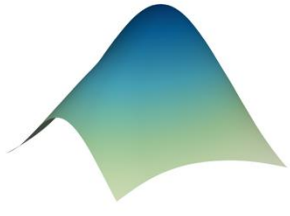
Diffusion Models



Diffusion Models

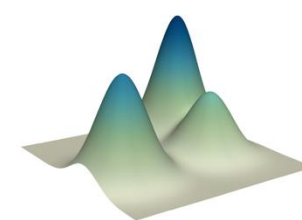
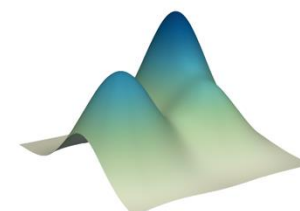
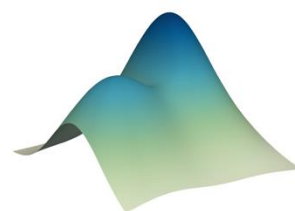
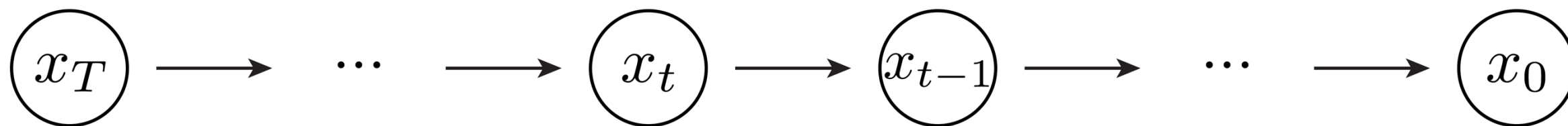


noise
distribution

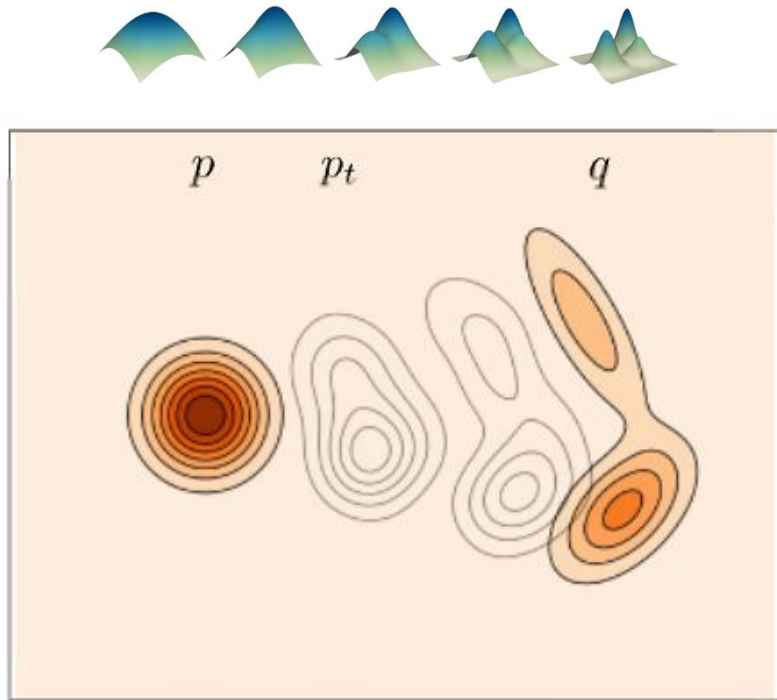


data
distribution

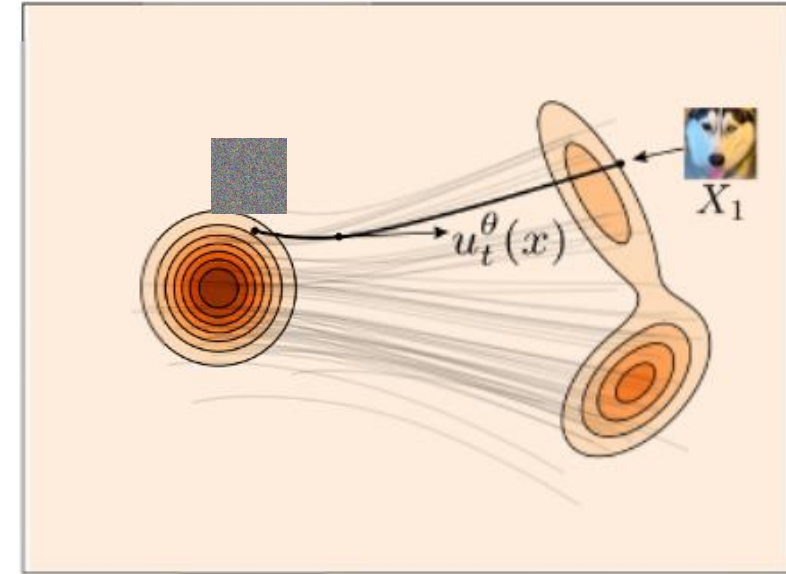
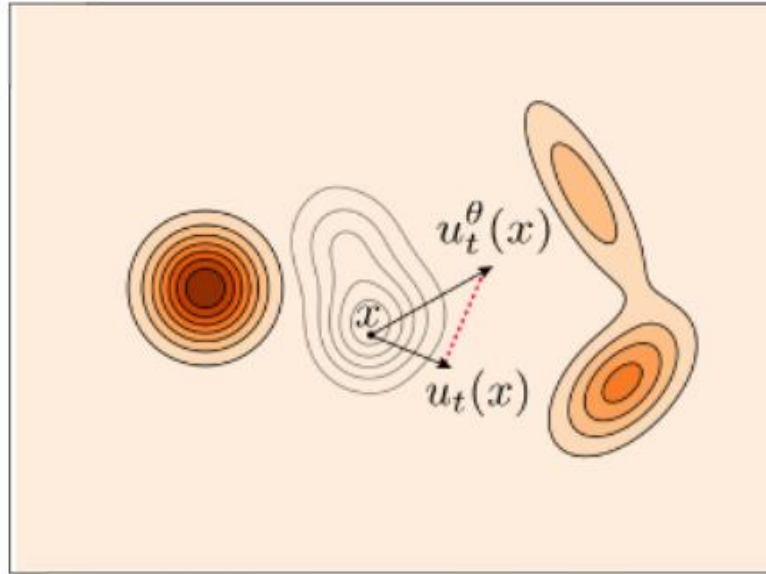
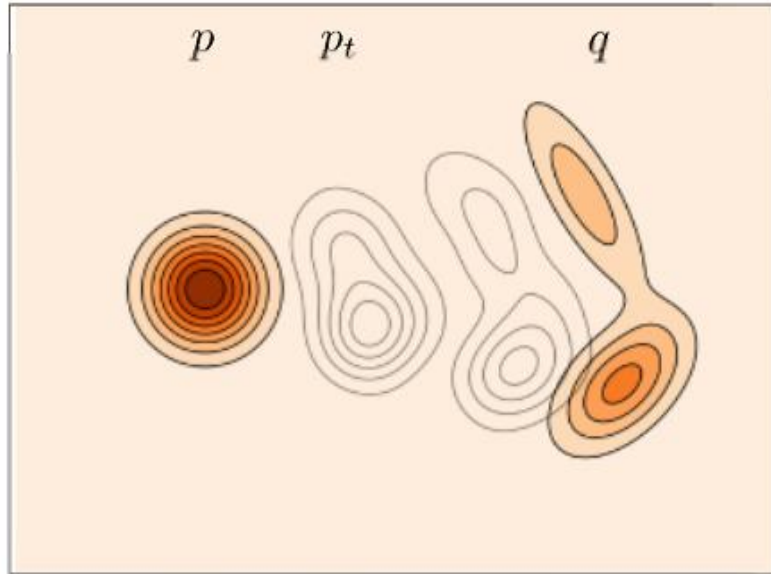
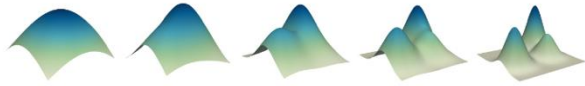
Diffusion Models



Flow Matching



Flow Matching



Takeaways

- Generative modeling is probabilistic modeling
- Learning to represent probabilistic distributions
- A way of problem-solving

This Lecture

- What are Generative Models?
- Generative Modeling of Real-world Problems
- Families of Generative Models